**VIDEO 359-1D Transthoracic echocardiographic images of a 9-year-old girl with first episode of acute rheumatic fever.** Images demonstrate the typical echocardiographic findings of acute rheumatic carditis. The valve leaflets are relatively thin and highly mobile. The failure of coaptation of the mitral valve leaflets is the result of chordal elongation and annular dilatation. The mitral valve regurgitation is moderate with a typical posterolaterally directed regurgitant jet of rheumatic carditis. ***D.*** Acute rheumatic carditis (parasternal long-axis view color Doppler echocardiogram).

**VIDEO 359-2A Transthoracic echocardiographic images are from a 5-year-old boy with chronic rheumatic heart disease with severe mitral valve regurgitation and moderate mitral valve stenosis.** Images demonstrate the typical echocardiographic findings in advanced chronic rheumatic heart disease. Both the anterior and posterior mitral valve leaflets are markedly thickened. During diastole, the motion of the anterior mitral valve leaflet tip is restricted with doming of the body of the leaflet toward the interventricular septum. This appearance is commonly described as a "hockey stick" or an "elbow" deformity. ***A.*** Chronic rheumatic heart disease (parasternal long-axis view).

**VIDEO 359-2B Transthoracic echocardiographic images are from a 5-year-old boy with chronic rheumatic heart disease with severe mitral valve regurgitation and moderate mitral valve stenosis.** Images demonstrate the typical echocardiographic findings in advanced chronic rheumatic heart disease. Both the anterior and posterior mitral valve leaflets are markedly thickened. During diastole, the motion of the anterior mitral valve leaflet tip is restricted with doming of the body of the leaflet toward the interventricular septum. This appearance is commonly described as a "hockey stick" or an "elbow" deformity. ***B.*** Chronic rheumatic heart disease (apical two-chamber view echocardiogram).

# 360 Systemic Sclerosis (Scleroderma) and Related Disorders

**John Varga**

## DEFINITION AND CLASSIFICATION

Systemic sclerosis (SSc) is an orphan disease of unknown etiology, complex pathogenesis, and variable clinical presentations. SSc frequently follows a progressive course and is associated with significant disability and mortality. Virtually every organ can be affected **(Fig. 360-1)**.

There is marked variability among SSc patients in patterns of skin involvement and organ complications, rates of disease progression, response to treatment, disease severity, and survival. The early stages of SSc are associated with prominent inflammatory features, but over time, structural alterations in multiple vascular beds and visceral organ dysfunction due to fibrosis and atrophy progressively dominate the clinical picture. Recently developed classification criteria for the diagnosis of SSc (shown in **Table 360-1**) are >90% specific and selective.

Although thickened and indurated skin (*scleroderma*) is the distinguishing hallmark of SSc, similar skin changes can also be seen in localized forms of scleroderma, along with a variety of metabolic, inherited, autoimmune, and iatrogenic conditions **(Table 360-2)**. Patients with SSc can be broadly segregated into two major subsets defined by the pattern of skin involvement and associated with characteristic clinical and serologic features and natural history **(Table 360-3)**. Patients with diffuse cutaneous SSc (dcSSc) have extensive skin induration, starting in the fingers (sclerodactyly) and ascending from distal to proximal limbs and the trunk. In these patients, progressive skin disease, interstitial lung disease (ILD), and less commonly acute renal involvement may develop relatively early. In contrast, in patients with limited cutaneous SSc (lcSSc), Raynaud's phenomenon generally precedes sclerodactyly and other disease manifestations, sometimes by years. In these patients, skin involvement remains confined to the fingers, distal limbs, and face, while the trunk is spared. A subset of patients with lcSSc display the characteristic constellation of clinical findings (calcinosis cutis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia), historically termed the *CREST syndrome*. In lcSSc, visceral organ involvement tends to follow an insidious and often benign course, while digital ischemic ulcers, pulmonary arterial hypertension (PAH), hypothyroidism, Sjogren's symptoms, and primary biliary cirrhosis may occur as late complications. In some patients, Raynaud's phenomenon and characteristic clinical and laboratory features of SSc occurs in the absence of detectable skin thickening. This relatively benign disease subset has been termed *SSc sine scleroderma.*

## INCIDENCE AND PREVALENCE

SSc is an acquired sporadic disease with a worldwide distribution and affecting all races. In the United States, the incidence is 9–46 cases per million per year. There are an estimated 100,000 U.S. cases, although this number may be significantly higher if patients who do not fulfill classification criteria are also included. There are large regional variations in incidence rates of SSc, potentially reflecting differences in case definition, environmental exposures, or genetic susceptibility genes in populations with different ancestries. Prevalence rates in England, northern Europe, and Japan appear to be lower than in North America and Australia. Age, sex, and ethnicity influence disease susceptibility, and blacks have higher age-specific incidence rates and mortality. In common with other connective tissue diseases, SSc shows a strong female predominance (4.6:1), which is most pronounced in the childbearing years and declines after menopause. An additional risk factor for SSc is having an affected first-degree family member, which increases disease risk 13-fold. Although SSc can present at any age, the peak age of onset in women with both limited and diffuse cutaneous forms is 65–74 years, although in blacks, disease onset occurs at an earlier age. Furthermore, blacks with SSc are more likely to have diffuse cutaneous disease, ILD, and a worse prognosis.

## GENETIC CONTRIBUTION TO DISEASE PATHOGENESIS

In general, the genetic associations of SSc identified to date make only a relatively modest contribution to disease susceptibility. Twin studies showed low disease concordance rates (4.7%) in monozygotic twins, a rate much lower than in other autoimmune diseases such as rheumatoid arthritis (12.3%). On the other hand, supporting the genetic contribution to disease susceptibility is the observation that 1.6% of SSc patients have a first-degree relative with SSc, a prevalence rate that is markedly increased compared to the general population. Moreover, the risk of Raynaud's phenomenon, ILD, and other autoimmune diseases, including systemic lupus erythematosus (SLE) **(Chap. 356)**, rheumatoid arthritis **(Chap. 358)**, and autoimmune thyroiditis **(Chap. 383)**, is also increased in first-degree relatives of patients with SSc. Current approaches to uncover the genetic factors that contribute to SSc include DNA sequencing and single nucleotide polymorphism (SNP) analysis of candidate genes and SNP analysis of the entire genome in a hypothesis-free manner. Genome-wide association studies (GWAS) involve large multicenter and multinational cohorts. A majority of the robustly validated genetic susceptibility loci for SSc are genes at the highly polymorphic human leukocyte antigen (HLA) region and other genes involved in innate and adaptive immunity and interferon responses, highlighting the importance of autoimmunity as the initial trigger for the disease. Genetic studies have shown associations with common (small effect size) variants related to B and T lymphocyte activation (*BANK1*, *BLK*, *CD247*, *STAT4*, *IL2RA*, *CCR6*, *IDO1*, *TNFSF4/OX40L*, *PTPN22*, and *TNIP1*). In addition, candidate gene studies and GWAS identified a strong and consistent association with HLA class II haplotypes on chromosome 6, including *HLA-DRB1*11:04*, *DQA1*05:01*, and *DQB1*03:01*, and the non-HLA histocompatibility complex (MHC) genes *NOTCH4* and *PSORSC1*. Other genetic variants associated with SSc are involved in innate immunity and type 1 interferon signaling (*IRF5*, *IRF7*, *STAT4*, *TNFAIP3*, and *TLR2*). Additional associations with *IL12RB2, IL-21,* the autophagy and apoptosis-related genes *DNASE1L3* and *SOX5*, and the

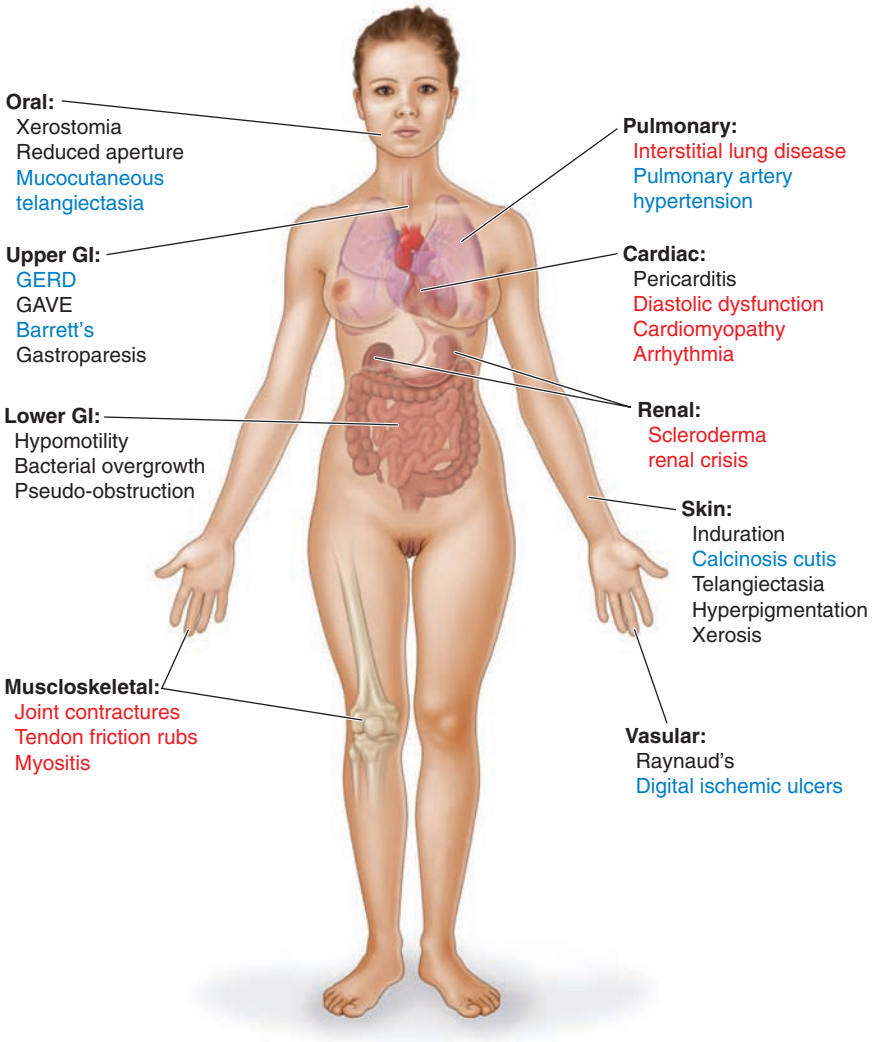



**FIGURE 360-1 Multiorgan involvement in systemic sclerosis (SSc).** Prominent complications of SSc: *red*, those more common in diffuse cutaneous SSc; *black*, those more common in limited cutaneous SSc; *blue*, complications common in both SSc subsets. GAVE, gastric antral vascular ectasia; GERD, gastroesophageal reflux disease; GI, gastrointestinal.

**TABLE 360-1 Classification Criteria for Diagnosis of Systemic Sclerosis**

| ITEM | SUBITEM | WEIGHT/SCORE |
|---|---|---|
| Skin thickening (bilateral); fingers extending proximal to MCP joints | | 9 |
| Skin thickening of fingers only | Puffy fingers | 2 |
| | Sclerodactyly (skin thickened distal to MCP joints) | 4 |
| Fingertip lesions | Digital tip ulcer | 2 |
| | Fingertip pitting scars | 3 |
| Mucocutaneous telangiectasia | | 2 |
| Abnormal nailfold capillary pattern | | 2 |
| Lung involvement | PAH | 2 |
| | Interstitial lung disease | 2 |
| Raynaud's phenomenon | | 3 |
| SSc-specific autoantibodies | ACA | 3 |
| | Scl-70 | |
| | RNA polymerase III | |

*Abbreviations:* ACA, anticentromere antibodies; MCP, metacarpophalangeal joint; PAH, pulmonary arterial hypertension.

**TABLE 360-2 Conditions Associated with Skin Induration**

- Systemic sclerosis (SSc)
  - Limited cutaneous SSc
  - Diffuse cutaneous SSc
- Localized scleroderma
  - Guttate (plaque) morphea, diffuse (panssclerotic) morphea, bullous morphea
  - Linear scleroderma, coup de sabre, hemifacial atrophy
- Panssclerotic morphea
- Overlap syndromes
  - Mixed connective tissue disease
  - SSc/polymyositis
- Diabetic scleredema and scleredema of Buschke
- Scleromyxedema (papular mucinosis)
- Chronic graft-versus-host disease
- Diffuse fasciitis with eosinophilia (Shulman's disease, eosinophilic fasciitis)
- Stiff skin syndrome
- Pachydermoperiostosis (primary hypertrophic osteoarthropathy)
- Chemically induced and drug-associated scleroderma-like conditions
  - Vinyl chloride–induced disease
  - Eosinophilia-myalgia syndrome (associated with L-tryptophan contaminant exposure)
  - Nephrogenic systemic fibrosis (associated with gadolinium exposure)
- Paraneoplastic syndrome

**TABLE 360-3 Subsets of Systemic Sclerosis (SSc): Features of Limited Cutaneous Versus Diffuse Cutaneous Disease**

| CHARACTERISTIC FEATURE | LIMITED CUTANEOUS SSc | DIFFUSE CUTANEOUS SSc |
|---|---|---|
| Skin involvement | Indolent onset. Limited to fingers, distal to elbows, face; slow progression | Rapid onset. Diffuse: fingers, extremities, face, trunk; rapid progression |
| Raynaud's phenomenon | Antedates skin involvement, sometimes by years; may be associated with critical ischemia in the digits | Onset coincident with skin involvement; critical ischemia less common |
| Musculoskeletal | Mild arthralgia | Severe arthralgia, carpal tunnel syndrome, tendon friction rubs |
| Interstitial lung disease | Slowly progressive, generally mild | Frequent, early onset and progression, can be severe |
| Pulmonary arterial hypertension | Frequent, late, may occur as an isolated complication | Often occurs in association with interstitial lung disease |
| Scleroderma renal crisis | Very rare | Occurs in 15%; generally early (<4 years from disease onset) |
| Calcinosis cutis | Frequent, prominent | Less common, mild |
| Characteristic autoantibodies | Anti-centromere | Anti–topoisomerase I (Scl-70), anti-RNA polymerase III |

extracellular matrix–related genes *CSK*, *CAV1*, *PPARG*, and *GRB10* have been reported. In addition to SSc susceptibility, some of these genetic loci are associated with particular disease manifestations or serologic subsets, including ILD (*CTGF*, *CD226*), PAH (*TNIP1*), scleroderma renal crisis (*HLA-DRB1**), and anticentromere antibodies (*HLA-DPB1*05:01*). While the functional consequences of these gene variants and their potential roles in pathogenesis are not well understood, it seems likely that in combination they cause a state of altered immune regulation, leading to increased susceptibility to autoimmunity and persistent inflammation. Of note, several of the SSc genetic variants are also implicated in other autoimmune disorders, including SLE, Sjögren's syndrome, rheumatoid arthritis, multiple sclerosis, and psoriasis, suggesting common pathogenic pathways shared among these phenotypically dissimilar conditions. Notably, the genetic associations identified to date only explain a fraction of the heritability of SSc and focus on common variants. By contrast, next-generation sequencing, such as whole exome sequencing, may help identify additional genetic susceptibility factors in SSc, particularly rare (and potentially causal) coding variants and their association with specific phenotypes.

## ENVIRONMENTAL AND OCCUPATIONAL EXPOSURES

The etiology of SSc is unknown. Given the relatively modest genetic contribution to susceptibility in SSc, environmental factors, such as infectious agents, microbiome, and occupational, dietary, lifestyle, and drug exposures, are likely to play a major role. Evidence suggests potential pathogenic roles for viruses, including parvovirus B19, Epstein-Barr virus (EBV), and cytomegalovirus (CMV), and *Rhodotorula glutinis*. Toxic oil syndrome, a novel disease with features suggestive of SSc, occurred as an epidemic outbreak in Spain in the 1980s and was linked to contaminated rapeseed oil used for cooking. Another epidemic outbreak, termed *eosinophilia-myalgia syndrome* (EMS), occurred in the United States in the 1990s and was linked to consumption of L-tryptophan-containing dietary supplements. Exposure to gadolinium contrast material in individuals with compromised renal function undergoing magnetic resonance scanning has been associated with nephrogenic systemic fibrosis. While each of these novel toxic-epidemic syndromes was characterized by chronic indurative skin changes and variable visceral organ involvement, the constellation of associated clinical, pathologic, and laboratory features distinguishes them from SSc. Occupational exposures tentatively linked with SSc include particulate silica (quartz), polyvinyl chloride, epoxy resins, welding fumes, and organic solvents and aromatic hydrocarbons including paint thinners, toluene, xylene, and trichloroethylene. These exposures can elicit cell type–specific stable and heritable epigenetic modifications such as DNA methylation and histone modification that could drive pathogenic alterations in cellular gene expression. Several of these epigenetic modifications are reversible and represent potential targets for therapy. Drugs implicated in SSc-like illnesses include bleomycin, pentazocine, cocaine, and appetite suppressants linked with PAH. Most recently, immune checkpoint inhibitors such as PD-1 blockers used in cancer therapy have been implicated as triggers for SSc-like illnesses. Radiation therapy for cancer has been linked with de novo onset of SSc as well as with exacerbation of preexisting SSc. In contrast to rheumatoid arthritis, cigarette smoking does not increase the risk of SSc. Although case reports and series describing the occurrence of SSc in women with silicone breast implants had raised concern regarding a putative pathogenic role of silicone in SSc, large-scale epidemiologic investigations found no evidence of increased prevalence of SSc.

## PATHOGENESIS

The pathogenesis of SSc involves putative environmental insults that trigger epigenetic modifications in a genetically predisposed host, causing alterations in gene expression underlying durable changes in the behavior of multiple cell types **(Fig. 360-2)**.

Three cardinal pathomechanistic processes underlie the protean clinical manifestations and pathologic changes of SSc: (1) diffuse microangiopathy, (2) inflammation and autoimmunity, and (3) visceral and vascular fibrosis affecting multiple organs **(Fig. 360-3)**. While each of these distinct processes may be synchronously active in a given SSc patient, their relative severity, progression, and contribution to the overall clinical picture vary among individual patients and over time. In general, autoimmunity and reversible vascular reactivity are early features of SSc, whereas fibrosis and atrophy occur later in the disease.

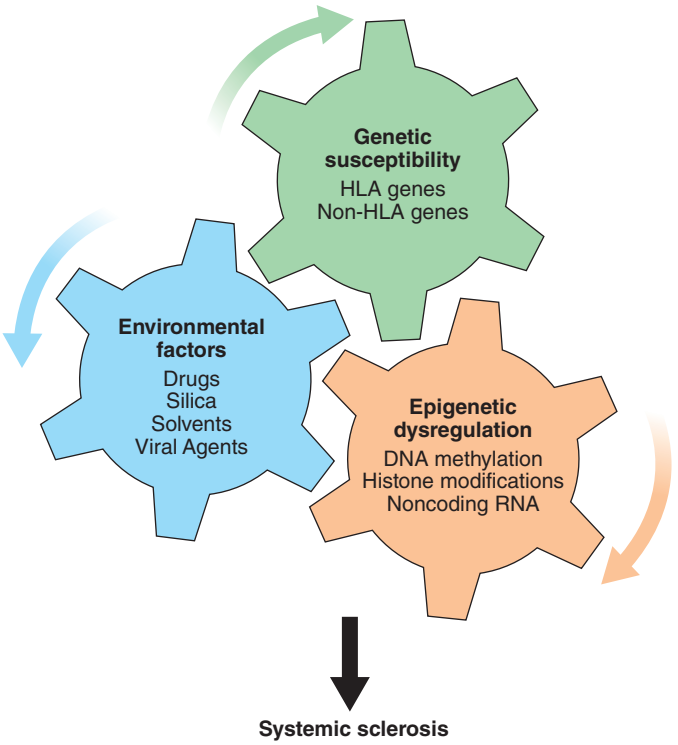



**FIGURE 360-2 The interplay of genetic risk factors and environmental exposure–induced epigenetic modifications underlying the complex pathogenesis of systemic sclerosis.** HLA, human leukocyte antigen. *(Reproduced with permission from Amr Sawalha and Pei-Suen Tsou, University of Pittsburgh and University of Michigan.)*

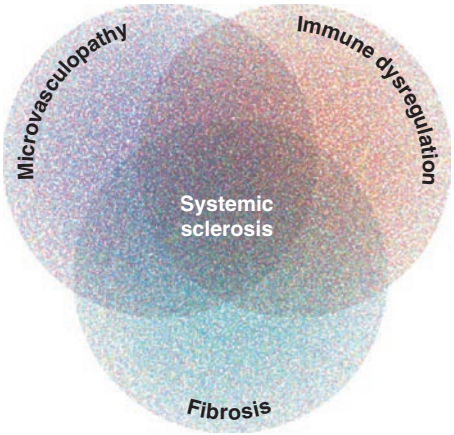



**FIGURE 360-3 The pathogenic systemic sclerosis (SSc) triad.** The characteristic constellation of synchronously occurring vasculopathy, autoimmunity/inflammation, and fibrosis distinguishes SSc and underlies its protean clinical manifestations.

## ANIMAL MODELS OF DISEASE

There is no single animal model of SSc that fully reproduces the three cardinal processes that underlie pathogenesis. Tight-skin mice (Tsk1/+) spontaneously develop skin (hypodermal) fibrosis due to a duplication mutation in the fibrillin-1 gene. Mutant fibrillin-1 protein disrupts extracellular matrix assembly, leading to aberrant activation of the profibrotic transforming growth factor β (TGF-β). In humans, fibrillin-1 mutations are associated with Marfan's disease and stiff skin syndrome but have not been reported in SSc. In mice, skin and lung fibrosis accompanied by variable vasculopathy and autoimmunity can be elicited by repeated injection of bleomycin or angiotensin II or by transplantation of HLA-mismatched bone marrow or spleen cells. Targeted genetic modifications in mice give rise to new disease models for investigating the pathogenetic roles of individual molecules, pathways, and cell types. For example, mice lacking IRF5, the ciliary proteins SPAG17, tenascin-C, or peroxisome proliferator-activated receptor (PPAR)-γ, or constitutively overexpressing β-catenin, Wnt10b, sirtuin 3, Fra2, PDGFRα, or adiponectin are either resistant or hypersensitive to experimental scleroderma or spontaneously develop multiple-organ fibrosis. These disease models can be useful as experimental tools to understand SSc pathogenesis and discover and validate novel targets for therapy.

## MICROANGIOPATHY

In a progressive model of disease pathogenesis **(Fig. 360-4)**, vascular injury is an early and possibly primary pathogenic event that underlies protean manifestations of small vessel vasculopathy.

Prominent microangiopathy in multiple vascular beds is a hallmark of SSc with important clinical sequelae including mucocutaneous telangiectasia, Raynaud's phenomenon, ischemic digital ulcers, scleroderma renal crisis, myocardial involvement, and PAH. Raynaud's phenomenon, commonly the initial manifestation of SSc, is characterized by altered blood-flow response to cold challenge in small digital arteries. This reversible functional abnormality is associated with autonomic and peripheral nervous system alterations, including impaired production of the neuropeptide calcitonin gene–related peptide from sensory afferent nerves and heightened sensitivity of $\alpha_2$-adrenergic receptors on vascular smooth-muscle cells. Isolated (primary) Raynaud's disease is common, generally benign, and nonprogressive. In contrast, SSc-associated secondary Raynaud's phenomenon often progresses to irreversible structural changes in the small blood vessels, culminating in ischemic digital tip ulcers, necrosis, and amputation.

Viruses, cytotoxic factors, chemokines, thrombogenic microparticles, alternate complement pathway activation, and autoantibodies targeting endothelial cells, phospholipids, and $\beta_2$-glycoprotein I ($\beta_2$GPI) have all been implicated as putative triggers of endothelial cell injury in SSc. Endothelial damage disrupts the production of vasodilatory (nitric oxide and prostacyclin) and vasoconstricting (endothelin-1) substances, while causing upregulation of intercellular adhesion molecule 1 (ICAM-1) and other surface adhesion molecules. Microvessels show enhanced permeability and transendothelial leukocyte diapedesis, activation of coagulation cascades, elevated thrombin production, and impaired fibrinolysis. Spontaneous platelet aggregation causes release of serotonin, platelet-derived growth factor (PDGF), and platelet alpha granules including thromboxane, a potent vasoconstrictor. Smooth-muscle cell–like myointimal cells accumulate in the media, potentially arising through a process called endothelial-mesenchymal transition (EndoMT). The basement membrane is thickened and reduplicated, and perivascular adventitial fibrosis develops. The vasculopathic process primarily affects capillaries, arterioles, and less commonly even large vessels in many organs, resulting in impaired blood flow and tissue ischemia. Progressive luminal occlusion due to intimal and medial hypertrophy, combined with persistent endothelial cell damage and adventitial fibrosis, establish a vicious cycle that underlies fibroproliferative vasculopathy and culminates in the striking absence of small blood vessels (rarefaction) in late-stage disease. Recurrent ischemia-reperfusion generates reactive oxygen species (ROS) that further damage the endothelium through peroxidation of membrane lipids. Paradoxically, the process of revascularization that normally reestablishes blood flow to ischemic tissue is defective in SSc despite elevated levels of other angiogenic factors. Moreover, bone marrow–derived circulating endothelial progenitor cells are reduced in number and impaired in function.

There is increasing evidence implicating EndoMT in the pathogenesis of SSc vasculopathy. The process of EndoMT is characterized by transition of endothelial cells into myofibroblasts accompanied by loss of endothelial cell markers and acquisition of myofibroblast markers associated with nuclear localization of the transcription factor Snail1. In arterioles and small arteries, EndoMT leads to accumulation of endothelial cell–derived myofibroblasts in the intima and media, resulting in fibroproliferative vasculopathy and luminal occlusion. In contrast, EndoMT affecting the capillary vessels leads to a destructive vasculopathy characterized by loss of endothelial cells and accumulation of interstitial myofibroblasts derived from endothelial cells, resulting in interstitial fibrosis and microvessel rarefaction, as can be observed by nailfold capillaroscopy. Widespread capillary loss combined with fibroproliferative vasculopathy affecting arterioles and arteries and impaired ability to repair and replace damaged vessels are hallmarks of SSc.

## INFLAMMATION AND AUTOIMMUNITY

**Cellular Immunity** While the initial events triggering the activation of innate and adaptive autoimmunity in SSc are unknown, a number of observations support the inflammatory/autoimmune nature of SSc: near-universal presence of circulating autoantibodies with defined specificities; clustering of SSc with other autoimmune diseases; activated immune cells, including autoreactive T cells with oligoclonal antigen receptors, within target organs; prominent type I interferon (IFN) signatures, characterized by elevated expression of IFN-regulated genes, in a variety of immune cell types in circulation and in skin biopsies; elevated circulating levels and spontaneous mononuclear cell secretion of cytokines and chemokines such as interleukin (IL) 6, tumor necrosis factor, IL-4, IL-10, IL-17, IL-33, CCL2, and CXCL4; genetic association of SSc (shared with other autoimmune diseases) with variants of MHC and other immune response genes; and the rapid clinical response, fibrosis resolution, and vascular regeneration observed in some SSc patients treated with immunomodulatory or immunoablative therapies.

Circulating monocytes from SSc patients overexpress IFN-regulated genes such as Siglec-1, have reduced levels of caveolin-1, and exhibit a profibrotic phenotype. In early (edematous) stage SSc, mononuclear cell infiltrates composed of activated T cells, monocytes/macrophages, and dendritic cells can be detected in skin, lungs, and other affected organs even prior to fibrosis or vascular damage. Dendritic cells in close proximity to activated fibroblasts and myofibroblasts express toll-like receptors (TLRs) that are activated by self-nucleic acids and other endogenous ligands. TLR stimulation induces the secretion

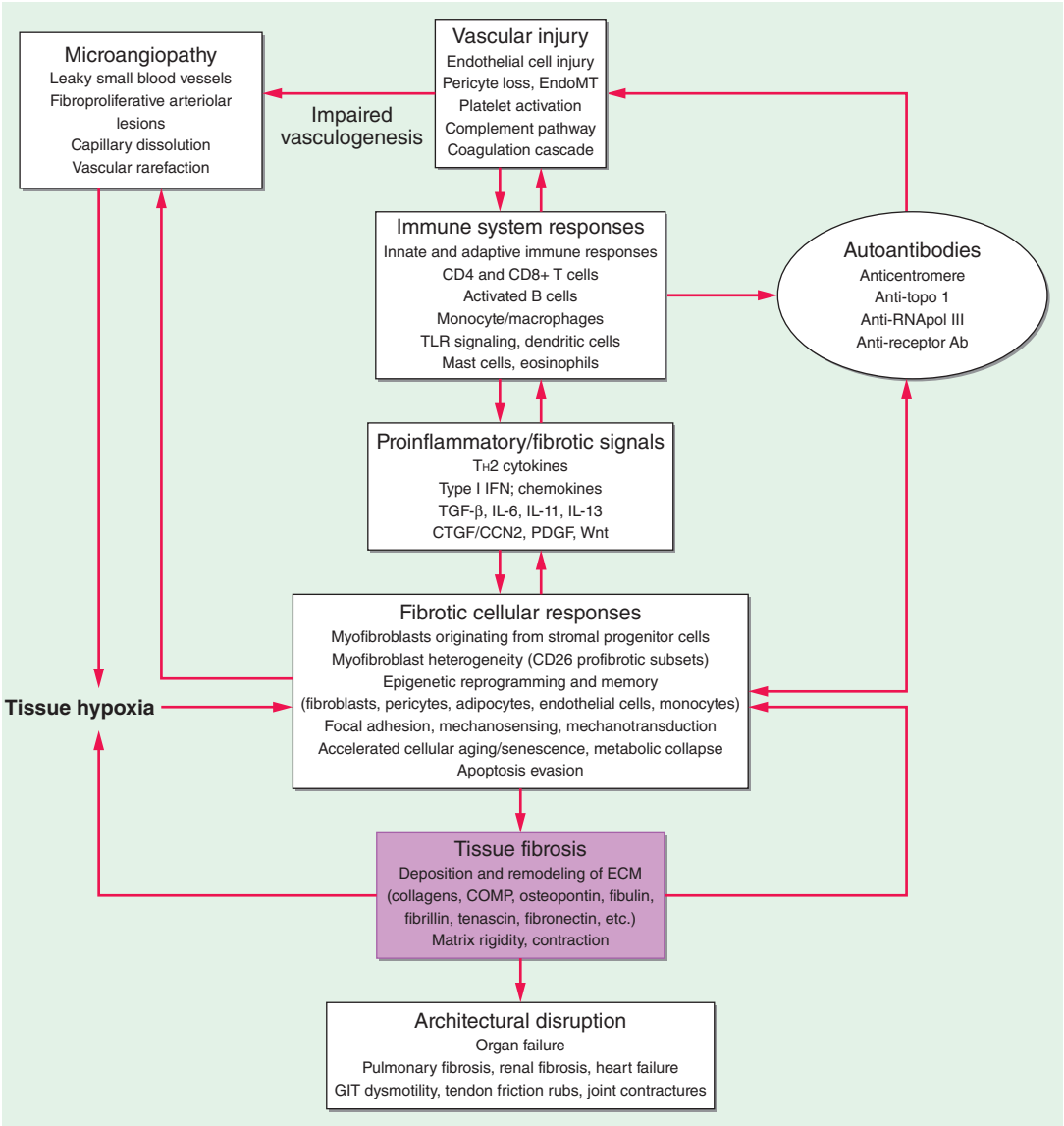



**FIGURE 360-4 Integrated progressive model of systemic sclerosis (SSc) pathogenesis.** Initial vascular insult in a genetically predisposed individual triggers a cascade of functional and structural vascular alterations associated with inflammation and autoimmunity. Early immune responses elicit fibroblast activation and differentiation, resulting in sustained pathologic fibrogenesis, irreversible tissue damage, and failure of affected organs. Vascular damage also leads to tissue ischemia that further contributes to progressive fibrosis and atrophy. Ab, antibody; CTGF, connective tissue growth factor; ECM, extracellular matrix; EndoMT, endothelial-mesenchymal transition; GIT, gastrointestinal tract; IFN, interferon; IL, interleukin; PDGF, platelet-derived growth factor; TGF-β, transforming growth factor β; TLR, toll-like receptor.

of mediators including IFN, IL-10, thymic stromal lymphopoietin (TSLP), and CXCL4, shaping the adaptive immune response and contributing to loss of immune tolerance. Tissue-infiltrating T cells express CD45 and HLA-DR activation markers and display restricted T-cell receptor signatures indicative of oligoclonal expansion in response to recognition of as-yet-unknown antigen. Of note, in patients diagnosed with SSc in close temporal association with cancer who are positive for RNA polymerase III antibody, the tumor commonly harbors genetic alterations in RNApol3, which results in the generation of autoantigen-specific T-cell immunity and cross-reactive antibodies.

Circulating T cells in SSc express chemokine receptors and $\alpha_1$ integrin, accounting for their enhanced binding to endothelium and to fibroblasts, while endothelial cells express ICAM-1 and other adhesion molecules that facilitate leukocyte diapedesis. Activated T cells show a $T_H2$-polarized immune response driven by dendritic cells. The $T_H2$ cytokines IL-4, IL-13, IL-33, and TSLP induce fibroblast activation, whereas the $T_H1$ cytokine interferon γ (IFN-γ) blocks cytokine-mediated fibroblast activation and exhibits antifibrotic properties. Evidence for altered $T_H17$ and regulatory T-cell function in SSc has been reported. Type 2 innate lymphoid cells (iLCs), a recently discovered distinct lymphoid cell population implicated in type 2 immunity and tissue remodeling, are also elevated in SSc skin biopsies. Alternately activated macrophages expressing CD163, which produce TGF-β and promote angiogenesis and tissue remodeling, are increased in the skin and lung in SSc. Regulatory T cells (Tregs) enforce immune tolerance, and although their frequency is elevated in the circulation and tissues in SSc patients, their immunosuppressive function appears to be defective. A recent report demonstrated that Tregs play a critical role in preventing spontaneous fibrosis in the skin, possibly by sequestering activated TGF-β. Some evidence implicates altered B-cell homeostasis and function in SSc. Circulating B cells show elevated CD19 and co-stimulatory molecules CD80 and CD86, suggesting B-cell chronic activation. Serum levels of a proliferation-inducing ligand (APRIL) and B-cell activating factor (BAFF), members of the TNF superfamily with potent effects on B-cell activation, are elevated in SSc and associated with extent of skin and lung involvement. B cells secrete IL-6, TGF-β, and other profibrotic cytokines implicated in pathogenesis. Thus, B-cell hyperactivity might directly contribute to the inflammatory and fibrotic processes in SSc, as well as generation of autoantibodies.

**Humoral Autoimmunity** Circulating antinuclear antibodies (ANAs) can be detected in virtually all patients with SSc, even in

**TABLE 360-4 Major Systemic Sclerosis (SSc)–Specific Autoantibodies and Principal Associated Features**

| TARGET ANTIGEN | SSc SUBSET | PROMINENT CHARACTERISTIC CLINICAL ASSOCIATION |
|---|---|---|
| Topoisomerase I (Scl-70)<br>Speckled pattern | dcSSc | Tendon friction rubs, digital ischemic ulcers, scleroderma, extensive skin involvement, early ILD, cardiac involvement, scleroderma renal crisis |
| Centromere proteins<br>Discrete speckled (centromere) pattern | lcSSc | Digital ischemic ulcers, calcinosis cutis, isolated PAH; renal crisis rare |
| RNA polymerase III<br>Speckled pattern | dcSSc | Rapidly progressive skin involvement, tendon friction rubs, joint contractures, GAVE, renal crisis, contemporaneous cancers; digital ulcers rare |
| U3-RNP (fibrillarin)<br>Nucleolar pattern | dc/lcSSc | PAH, ILD, scleroderma renal crisis, GI tract involvement, myositis |
| Th/$T_0$<br>Nucleolar pattern | lcSSc | ILD, PAH |
| PM/Scl<br>Nucleolar pattern | lcSSc | Calcinosis cutis, ILD, myositis overlap |
| Ku<br>Speckled pattern | Overlap | SLE, myositis overlap |
| U1-RNP<br>Speckled pattern | MCTD | PAH, inflammatory arthritis, myositis overlap |
| U11/U12 RNP<br>Speckled pattern | dc/lcSSc | ILD |

*Abbreviations:* dcSSc, diffuse cutaneous SSc; GAVE, gastric antral vascular ectasia; GI, gastrointestinal; ILD, interstitial lung disease; lcSSc, limited cutaneous SSc; MCTD, mixed connective tissue disease; PAH, pulmonary arterial hypertension; SLE, systemic lupus erythematosus.

early and possibly preclinical stages of disease. In addition, several SSc-specific autoantibodies with distinct patterns of immunofluorescence show strong associations with unique disease phenotypes as well as HLA haplotype (Table 360-4). Owing to their specificity, mutual exclusivity, and association with unique disease manifestations, SSc-associated autoantibodies have substantial utility in clinical practice for diagnosis and risk stratification, while their role in monitoring disease activity or response to therapy remains uncertain. Recently, antibodies directed against fibrillin-1, matrix metalloproteinases, cell surface markers, angiotensin II receptor, endothelin-1 receptor, or the PDGF receptor have been identified in patients with SSc, although their clinical relevance is not yet established. These antibodies have functional receptor agonist activity and might have direct pathogenic roles.

A variety of mechanisms have been proposed to account for the generation of SSc-associated autoantibodies. Proteolytic cleavage, increased expression or altered subcellular localization of normal proteins, or their alterations due to mutation in the case of certain tumors could lead to their immune recognition as neoepitopes, resulting in a break of immune tolerance.

## FIBROSIS

Fibrosis synchronously affecting multiple organs is a distinguishing feature of SSc. The process is characterized by replacement of normal tissue architecture with rigid, avascular, and relatively acellular connective tissue. Fibrosis in SSc follows, and is a consequence of, inflammation and microvascular damage (Fig. 360-4). Fibroblasts are mesenchymal cells primarily responsible for the functional and structural integrity of connective tissue. Upon their activation by extracellular cues, fibroblasts proliferate; migrate; secrete collagens and other matrix molecules, growth factors, chemokines, and cytokines; and transdifferentiate into contractile myofibroblasts. Under normal conditions, these are responses self-limited to accomplish tissue regulated repair and regeneration. In contrast, when these responses become sustained and amplified, pathologic fibrosis results. A panoply of stimulatory signals are potentially implicated in SSc pathogenesis. In addition to TGF-β, these include paracrine mediators IL-6, IL-11, IL-13, and IL-23, morphogens and Wnt ligands, connective tissue growth factor (CTGF), PDGF, lysophosphatidic acid, endothelin-1, hypoxia, ROS, thrombin, and mechanical forces; these signals might contribute to sustained fibroblast activation underlying maladaptive repair in SSc. Buildup of damage-associated endogenous ligands for TLR4 (EDA-fibronectin and tenascin-C) and TLR9 (mitochondrial DNA) within the microenvironment further contributes to nonresolving fibrosis via unchecked TLR activation and innate immune signaling.

In addition to tissue-resident fibroblasts and transformed myofibroblasts, bone marrow–derived circulating mesenchymal progenitor cells also contribute to fibrosis. The factors that regulate the differentiation of mesenchymal progenitor cells and their trafficking from the circulation into lesional tissue are unknown. Endothelial cells in injured arterioles and small arteries undergo EndoMT, giving rise to myofibroblasts that drive perivascular fibrosis. Epithelial cells, preadipocytes, and tissue fibroblasts are all putative sources of pathogenic myofibroblasts. Although myofibroblasts are transiently found in normal wound healing, their persistence in fibrotic tissue, possibly due to their ability to evade apoptosis, contributes to scar formation.

Explanted SSc fibroblasts display an abnormally activated phenotype ex vivo, characterized by increased collagen production, spontaneous ROS generation, prominent stress fibers, and constitutive expression of alpha smooth-muscle actin. Persistence of the "scleroderma phenotype" during serial ex vivo passage of SSc fibroblasts may reflect autocrine TGF-β stimulatory loops, deregulated microRNA expressions, cell-autonomous metabolic changes, or stable acquired epigenetic modifications. More recently, tools such as single-cell RNA sequencing have revealed functional heterogeneity among fibrotic fibroblasts.

# PATHOLOGY

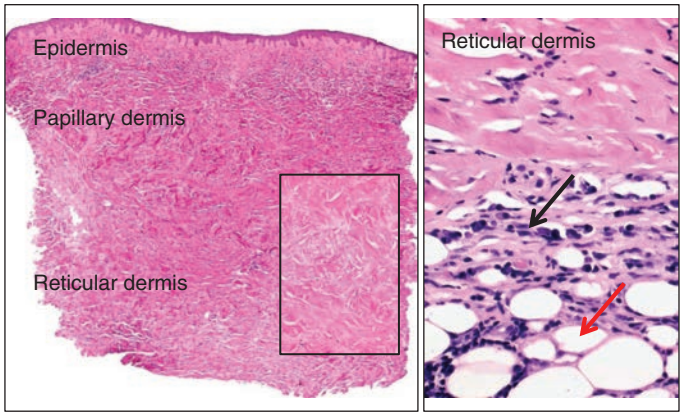
While pathologic findings vary across anatomic sites, the distinguishing hallmarks of SSc irrespective of the organ system are widespread microangiopathy (fibroproliferative vasculopathy), capillary loss and obliteration, and fibrosis. In early-stage disease, perivascular inflammatory cell infiltrates composed of T and B lymphocytes, activated monocytes and macrophages, and mast cells may be detected in multiple organs. A noninflammatory obliterative microangiopathy in the heart, lungs, kidneys, and gastrointestinal tract is a prominent late finding. Fibrosis most prominently affects the skin, lungs, cardiovascular system, gastrointestinal tract, tendon sheaths, perifascicular tissue surrounding skeletal muscle, and some endocrine organs such as the thyroid gland. Excessive accumulation of collagens and other structural matrix macromolecules progressively disrupts normal architecture, resulting in impaired function and failure of affected organs.

## SKIN

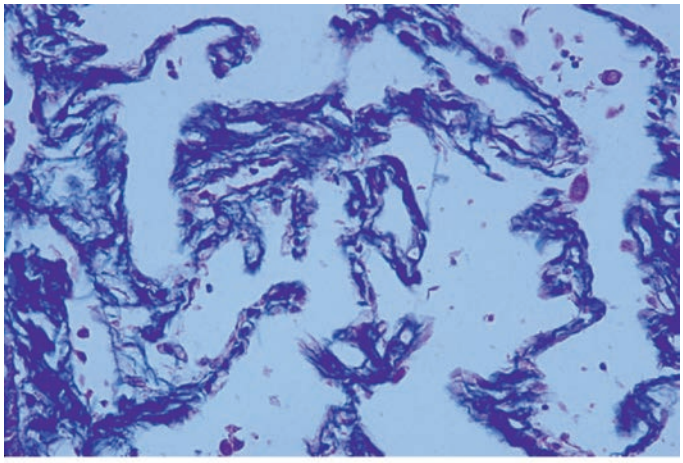
In the skin, the dermis is thickened due to accumulation of broad bundles of homogenized collagen oriented parallel to the epithelium (Fig. 360-5*A*). Adnexal glands are atrophic, and loss of periadnexal and intradermal white adipose tissue and its replacement with collagen can be striking. While perivascular mononuclear cell infiltrates may be seen early, established skin fibrosis shows absence of inflammation.

## LUNGS

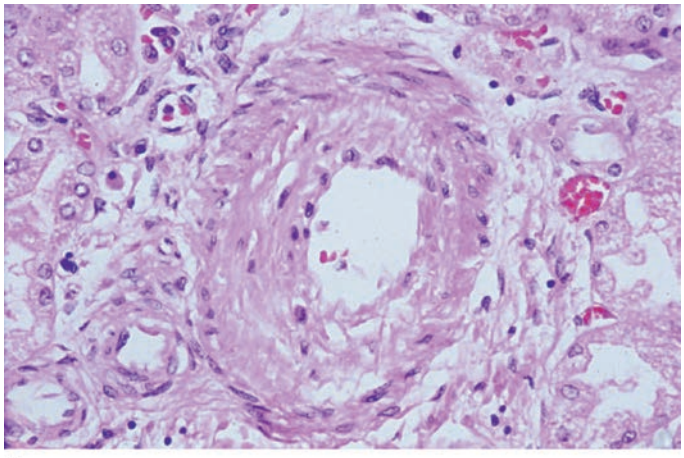
Autopsy studies universally show evidence of lung involvement in both limited and diffuse cutaneous subsets of SSc. Most common is the nonspecific interstitial pneumonia (NSIP) pattern characterized variable interstitial fibrosis and mild chronic inflammation with T lymphocytes, macrophages, and eosinophils. With progression, interstitial fibrosis and vascular damage dominate, often coexisting within the same biopsy. The usual interstitial pneumonia (UIP) pattern of spatial and temporal heterogeneity of inflammation and fibrosis and fibrotic foci, a hallmark of idiopathic pulmonary fibrosis, is less common in SSc (Fig. 360-5*B*). Fibrosis of the alveolar septae results in obliteration of the airspaces and loss of pulmonary blood vessels. This process impairs gas exchange and contributes to pulmonary hypertension. Intimal




*A*



*B*



*C*

**FIGURE 360-5 Pathologic findings in multiple organs in systemic sclerosis (SSc).** ***A. Left panel:*** The skin is thickened due to fibrotic expansion of the dermis. Inset, higher magnification showing thick hyalinized collagen bundles replacing skin appendages. ***Right panel:*** Mononuclear inflammatory cells in the dermis and intradermal adipose tissue. ***B.*** Early SSc interstitial lung disease. Diffuse fibrosis of the alveolar septae and a chronic inflammatory cell infiltrate. Trichrome stain. ***C.*** Pulmonary arterial obliterative vasculopathy. Striking intimal hyperplasia and luminal narrowing of small artery, with little inflammation and minimal interstitial lung fibrosis, in a patient with SSc pulmonary arterial hypertension.

thickening of the pulmonary arteries, best seen with elastin stain, underlies SSc-associated PAH (Fig. 360-5*C*) and, at autopsy, is often associated with pulmonary emboli and myocardial fibrosis. Patients with SSc-associated PAH also show fibrosis and intimal proliferation in preseptal venules and veins in the lung, accounting for veno-occlusive disease. Lymphocytic bronchiolitis involving the submucosa of the terminal bronchioles is occasionally seen.

### GASTROINTESTINAL TRACT

Pathologic changes can be found at any level from the mouth to the rectum. Atrophy and fibrosis of the muscularis propria and characteristic vascular lesions are prominent in the lower esophagus, while striated muscle in the upper third of the esophagus is generally spared. Collagenous replacement of the intestinal tract architecture results in impaired smooth-muscle contractility and diminished peristaltic activity, underlying gut dysmotility, bacterial overgrowth, small-bowel pseudo-obstruction, and perforation. Chronic gastroesophageal reflux is associated with esophageal inflammation, mucosal ulceration, and stricture formation and may lead to Barrett's metaplasia with attendant risk of adenocarcinoma. Esophageal dilatation and reflux may aggravate ILD due to chronic microaspiration.

### KIDNEYS

In the kidneys, vascular lesions affecting the interlobular and arcuate arteries predominate. Chronic renal ischemia is associated with shrunken glomeruli. Patients with scleroderma renal crisis, a life-threatening acute complication of SSc, show acute fibrinoid necrosis of afferent arterioles, followed by intimal proliferation (onion-skin pattern) and ischemic collapse of glomeruli. These changes are reminiscent of thrombotic microangiopathies such as atypical hemolytic-uremic syndrome (Chap. 115) and are accompanied by thrombosis, thrombocytopenia due to platelet consumption, and intravascular hemolysis. Evidence of complement activation may be seen in kidney biopsies. Extensive vascular thrombosis, glomerular collapse, and peritubular capillary deposits predict irreversible renal failure.

### HEART

Subclinical cardiac pathology is common in SSc and may affect the myocardium and pericardium. The characteristic arteriolar lesions in the heart are concentric intimal hypertrophy and luminal narrowing, patchy contraction band necrosis, loss of cardiac myocytes, and myocardial fibrosis due to microvascular involvement and ischemia-reperfusion injury. Fibrosis of the conduction system is also common, especially at the sinoatrial node. The frequency of epicardial atherosclerotic coronary artery disease may be increased in SSc compared to the general population, similar to other systemic inflammatory diseases. Pericardial involvement with chronic inflammatory infiltrates and fibrinous exudates is common and is sometimes associated with pericardial effusions.

### PATHOLOGY IN OTHER ORGANS

Synovitis of the hands may occur in SSc; with disease progression, the synovium becomes fibrotic, and in contrast to rheumatoid disease, pannus formation and bone resorption are uncommon. Fibrosis of tendon sheaths and fascia, sometimes accompanied by calcifications, produces palpable and sometimes audible tendon friction rubs. Inflammation and, in later stages, atrophy and fibrosis of skeletal muscles are common findings and are similar to those in polymyositis. Fibrosis of the thyroid and of the minor salivary glands may be seen and underlie hypothyroidism and the sicca syndrome. Placentas from SSc pregnancies show decidual vasculopathy, which is associated with poor perinatal outcomes and fetal death.

## CLINICAL FEATURES

### OVERVIEW

SSc is truly a systemic disease that can affect virtually any organ (Fig. 360-1 and Table 360-5). Although a dichotomous stratification into diffuse and limited cutaneous subsets (Table 360-2) is useful, SSc is far more complex, and multiple distinct clusters or endophenotypes of SSc with characteristic manifestations and trajectories and outcomes can be recognized within each subset. Unique endophenotypes associate with autoantibodies with distinct and mutually exclusive specificities (Table 360-4). Moreover, patients with "overlap" have typical features of SSc coexisting with clinical and laboratory evidence of another autoimmune disease, most commonly polymyositis, Sjögren's syndrome, polyarthritis, autoimmune liver disease, or SLE.

**TABLE 360-5 Frequency of Clinical Organ Involvement in Limited Cutaneous and Diffuse Cutaneous Systemic Sclerosis (SSc)**

| FEATURES | LIMITED CUTANEOUS SSc (%) | DIFFUSE CUTANEOUS SSc (%) |
|---|---|---|
| Skin involvement | 90[a] | 100 |
| Raynaud's phenomenon | 99 | 98 |
| Ischemic digital ulcers | 50 | 25 |
| Esophageal involvement | 90 | 80 |
| Interstitial lung disease | 35 | 65 |
| Pulmonary arterial hypertension | 15 | 15 |
| Myopathy | 11 | 23 |
| Clinical cardiac involvement | 9 | 12 |
| Scleroderma renal crisis | 2 | 15 |
| Calcinosis cutis | 40 | 35 |

[a]Approximately 10% of patients have SSc *sine* scleroderma.



**FIGURE 360-6 Digital necrosis.** Sharply demarcated necrosis of the fingertip secondary to ischemia in a patient with limited cutaneous systemic sclerosis (SSc) associated with severe Raynaud's phenomenon.

### INITIAL CLINICAL PRESENTATION

Characteristic initial presentation is quite different in patients with the diffuse (dcSSc) versus limited (lcSSc) cutaneous forms of the disease. In dcSSc, the interval between Raynaud's phenomenon and onset of other disease manifestations is typically brief (weeks to months). Soft tissue swelling, puffy fingers, and pruritus are signs of the early inflammatory "edematous" phase. The fingers, distal limbs, and face are usually affected first. Diffuse hyperpigmentation of the skin, carpal tunnel syndrome, arthralgias, muscle weakness, fatigue, and decreased joint mobility are common. During the ensuing weeks to months, the inflammatory edematous phase evolves into the "fibrotic" phase, with skin induration associated with hair loss, reduced production of skin oils, and decline in sweating capacity. Progressive flexion contractures of the fingers ensue. The wrists, elbows, knees, and ankles become stiff due to fibrosis of the supporting joint structures. While advancing skin involvement is the most visible manifestation of early dcSSc, important and clinically silent internal organ involvement can occur during this stage. The initial 4 years from disease onset is the period of most rapidly evolving and potentially irreversible lung and renal damage. If organ failure does not occur during this phase of dcSSc, the systemic process may plateau and stabilize.

Compared to dcSSc, the course of lcSSc tends to be more indolent. The interval between onset of Raynaud's phenomenon and disease manifestations such as gastroesophageal reflux disease (GERD), cutaneous telangiectasia, ischemic digital ulcers, or soft tissue calcifications can be as long as years. Scleroderma renal crisis, significant ILD, and tendon friction rubs occur rarely in lcSSc, whereas PAH and overlap with keratoconjunctivitis sicca, polyarthritis, cutaneous vasculitis, and biliary cirrhosis can develop even many years after disease onset.

## ORGAN INVOLVEMENT

### RAYNAUD'S PHENOMENON

Raynaud's phenomenon, the most frequent extracutaneous complication of SSc, is characterized by episodic vasoconstriction in the fingers and toes, sometimes also affecting the tip of the nose and earlobes. Attacks are reversible, and can be triggered by a decrease in temperature, as well as emotional stress and vibration. Attacks typically start with pallor of the fingers, followed by cyanosis of variable duration. Hyperemia ensues spontaneously or with rewarming of the digit. The progression of the three color phases reflects the underlying vasoconstriction, ischemia, and reperfusion. It is important to note that up to 5% of the general population has Raynaud's phenomenon. In the absence of signs or symptoms of an underlying condition, Raynaud's phenomenon is classified as primary (Raynaud's disease), which represents an exaggerated physiologic vasomotor response to cold. Secondary Raynaud's phenomenon occurs in SSc and other connective tissue diseases, hematologic and endocrine conditions, and occupational disorders, and can complicate treatment with beta blockers and anticancer drugs such as cisplatin and bleomycin. Distinguishing primary Raynaud's disease from secondary Raynaud's phenomenon can present a diagnostic challenge. Raynaud's disease is supported by the following: absence of an underlying cause; a family history of Raynaud's phenomenon; absence of digital tissue necrosis or ulceration; and a negative ANA test. Secondary Raynaud's phenomenon tends to occur at an older age, is more severe (episodes are more frequent, prolonged, and painful), and is frequently complicated by ischemic digital ulcers and loss of digits (Fig. 360-6).

Nailfold capillaroscopy using a low-power stereoscopic microscope or ophthalmoscope permits visualization of nailbed cutaneous capillaries under immersion oil (Fig. 360-7). Raynaud's disease is associated with evenly spaced parallel vascular loops, whereas in secondary Raynaud's phenomenon, nailfold capillaries are distorted with widened and irregular loops, dilated lumen, microhemorrhages, and areas of vascular "dropout." Thus, nailfold capillaroscopy can be helpful for both differentiating primary from secondary Raynaud's phenomenon and for establishing the early diagnosis of SSc.

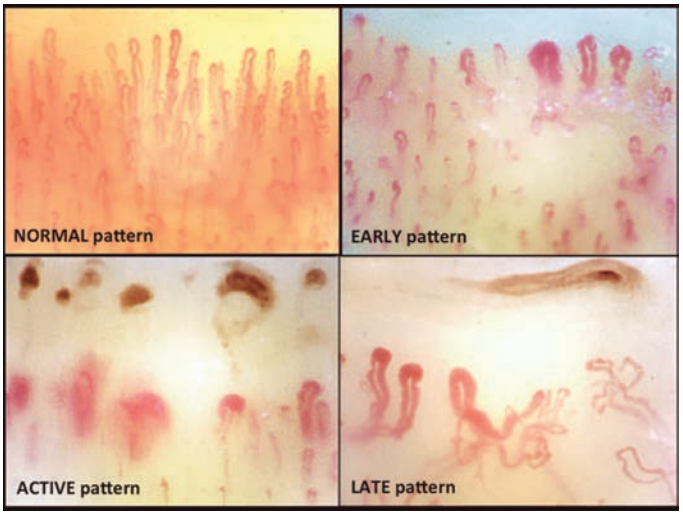



**FIGURE 360-7 Systemic sclerosis–associated nailfold capillary alterations.** In healthy subjects, note regularly arrayed and uniform-size "hairpin" microvessels. In early pattern, note dilations of microvessels and symmetrically increased microvessels (giant capillaries). In active pattern, note giant capillaries, collapse with microhemorrhages, and loss of capillaries. In late pattern, note massive loss of capillaries, fibrosis, and neoangiogenesis with secondary dilations (nailfold videocapillaroscopy; magnification 220×). *(Courtesy Professor Maurizio Cutolo, University of Genoa, Italy.)*

## SKIN FEATURES

Bilateral symmetrical skin thickening is the hallmark that distinguishes SSc from other connective tissue diseases. Skin involvement starts in the fingers and characteristically advances from distal to proximal extremities in an ascending fashion. Some patients note diffuse tanning in the absence of sun exposure as a very early manifestation. In dark-skinned individuals, vitiligo-like hypopigmentation may occur. Pigment loss sparing the perifollicular areas gives rise to a "salt-and-pepper" appearance of the skin, most prominently on the scalp, upper back, and chest. Dermal sclerosis obliterating hair follicles, sweat glands, and eccrine and sebaceous glands causes hair loss, decreased sweating, and xerosis and itching in affected areas of the skin. Transverse creases on the dorsum of the fingers disappear **(Fig. 360-8)**. Fixed flexion contractures of the fingers cause reduced hand mobility and lead to muscle atrophy. Skin and subjacent tendon fibrosis accounts for fixed contractures of the wrists, elbows, and knees. Thick ridges at the neck due to firm adherence of skin to the underlying platysma muscle interfere with neck extension.

Patients with established SSc may show a characteristic "mauskopf" facial appearance with taut and shiny skin, loss of wrinkles, and occasionally an expressionless facies due to reduced mobility of the eyelids, cheeks, and mouth. Thinning of the lips with accentuation of the central incisor teeth and prominent perioral radial furrowing (rhytides) complete the picture. Reduced oral aperture (microstomia) interferes with eating and oral hygiene. The nose assumes a pinched, beak-like appearance. In late-stage disease, the skin becomes thin and atrophic and is firmly bound to the subcutaneous fat (tethering). Dilated skin capillaries 2–20 mm in diameter (telangiectasia), reminiscent of hereditary hemorrhagic telangiectasia, are frequently on the face, hands, lips, and oral mucosa **(Fig. 360-9)**. The number of telangiectasias correlates with the severity of microvascular disease, including PAH. Breakdown of atrophic skin leads to chronic ulcerations at the extensor surfaces of the proximal interphalangeal joints, the volar pads of the fingertips, and bony prominences such as elbows and malleoli. Ulcers are often painful, heal slowly, and become secondarily infected, resulting in osteomyelitis. Healing of ischemic fingertip ulcerations leaves characteristic fixed digital "pits." Loss of soft tissue at the fingertips due to ischemia may be associated with striking resorption of the terminal phalanges (acro-osteolysis) **(Fig. 360-10)**.

Dystrophic calcifications in the skin, subcutaneous, and soft tissues (calcinosis cutis) in the presence of normal serum calcium and phosphate levels occur in up to 40% of patients, most commonly in those with long-standing anti-centromere antibody–positive lcSSc. Calcific deposits, composed of calcium hydroxyapatite crystals, vary in size from tiny punctate lesions to large conglomerate masses, and can be readily visualized on plain radiographs or dual-energy CT. These deposits occur when calcium precipitates in tissue damaged by inflammation, hypoxia, or local trauma. Common locations include the finger pads, palms, extensor surfaces of the forearms, and the olecranon and prepatellar bursae **(Fig. 360-11)**. Large calcific deposits can cause pain and nerve compression and may ulcerate through the



*A*



*B*

**FIGURE 360-9 Cutaneous vascular changes in systemic sclerosis.** ***A.*** Vascular changes at the nailfold in a patient with lcSSc. ***B.*** Telangiectasia on the face.



**FIGURE 360-8 Sclerodactyly.** Note skin induration on the fingers and fixed flexion contractures of proximal interphalangeal joints in a patient with limited cutaneous systemic sclerosis (lcSSc).

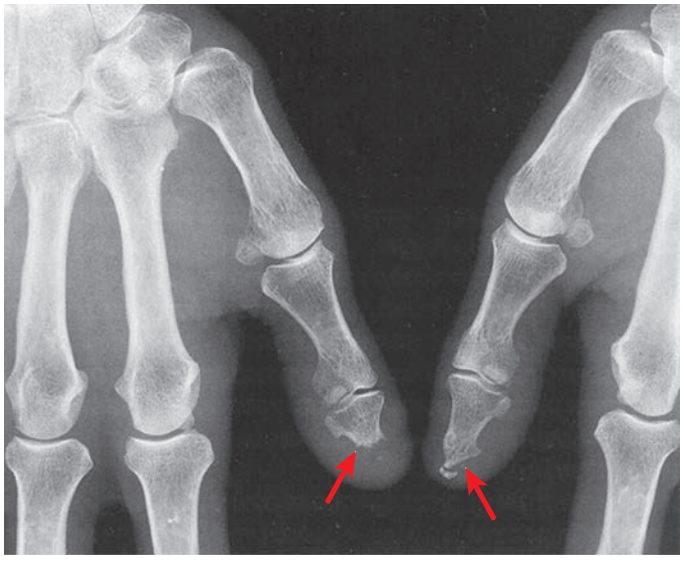


**FIGURE 360-10 Acro-osteolysis.** Note dissolution of distal terminal phalanges (*arrows*). Acro-osteolysis is associated with digital ischemia and is seen in patients with long-standing limited cutaneous systemic sclerosis (lcSSc) and Raynaud's phenomenon.



*A*



*B*

**FIGURE 360-11 Calcinosis cutis in systemic sclerosis.** ***A.*** Note calcific deposit breaking through the skin in a patient with limited cutaneous systemic sclerosis (lcSSc). ***B.*** Dual-energy computed tomography showing calcific deposits at the proximal interphalangeal joints.

overlying skin with drainage of chalky white material and secondary infections. Paraspinal sheet calcifications may cause neurologic complications.

## PULMONARY FEATURES

The two principal forms of lung involvement in SSc, ILD and pulmonary vascular disease, are frequent and together account for a majority of SSc-related deaths. Survival is particularly poor in SSc patients with concurrent presence of these two processes. Less common pulmonary complications of SSc include aspiration pneumonitis complicating chronic gastroesophageal reflux, pulmonary hemorrhage due to endobronchial telangiectasia, obliterative bronchiolitis, pleural reactions, restrictive physiology due to chest wall fibrosis, spontaneous pneumothorax, and drug-induced lung toxicity. The incidence of lung cancer is increased in SSc.

**Interstitial Lung Disease** While evidence of ILD can be found in up to 65% of SSc patients by high-resolution computed tomography (HRCT), clinically significant ILD develops in 16–43%; the frequency varies depending on the detection method used. Risk factors for significant ILD include male sex, African-American race, diffuse skin involvement, severe gastroesophageal reflux, and presence of topoisomerase-I (Scl-70) autoantibodies, whereas anti-centromere antibody–positive patients have a reduced risk of ILD. Additional risk factors for significant ILD include low forced vital capacity (FVC) or single-breath diffusing capacity of the lung for carbon monoxide ($DL_{CO}$) at initial presentation. Esophageal dilatation with chronic acid reflux in SSc causes recurrent micro-aspiration, a risk factor for the development and progression of ILD. The most rapid progression in ILD generally occurs early in the disease course (within the first 3–5 years), when the FVC can decline by 30% per year. In contrast, new onset of ILD is rare in SSc patients with long-standing disease.

Pulmonary involvement can remain asymptomatic until it is advanced. The most common presenting respiratory symptoms—exertional dyspnea, fatigue, and reduced exercise tolerance—are subtle and slowly progressive. A chronic dry cough may be present. Physical examination may reveal fine inspiratory "Velcro" crackles at the lung bases. Pulmonary function testing (PFT) is relatively insensitive for detecting early pulmonary involvement. In patients with established SSc-ILD, PFT typically shows a restrictive ventilatory defect (FVC <70% predicted and/or forced expiratory volume in 1 s [$FEV_1$]/FVC ratio >0.8) and reduced total lung capacity (TLC) and diffusing capacity ($DL_{CO}$). A reduction in $DL_{CO}$ that is significantly out of proportion to the reduction in lung volumes should raise suspicion for pulmonary vascular disease but may also be due to anemia. Oxygen desaturation with exercise is common.

Chest radiography can be used as an initial screening tool to rule out infection and other causes of respiratory symptoms in SSc, but compared to HRCT, it is relatively insensitive for detection of early ILD. Characteristic imaging findings include lower lobe subpleural reticular linear opacities and ground-glass opacifications with an apicobasal gradient, even in asymptomatic patients with normal PFTs **(Fig. 360-12)**. Additional HRCT findings include mediastinal lymphadenopathy, pulmonary nodules, traction bronchiectasis, and uncommonly, honeycomb changes. The extent of interstitial changes on chest HRCT at baseline is a predictor of ILD progression and mortality. While bronchoalveolar lavage (BAL) can demonstrate inflammatory cells in the lower respiratory tract and may be useful for ruling out tuberculosis and other infections, it does not appear to be useful for SSc diagnosis or for identifying reversible alveolitis. Surgical lung biopsy in SSc is

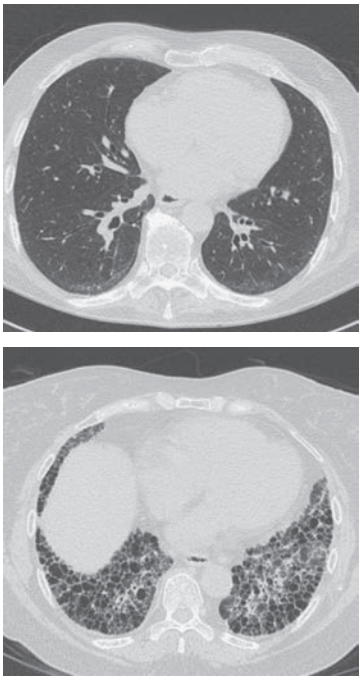


**FIGURE 360-12 High-resolution chest CT findings in systemic sclerosis.** ***Top panel:*** Early interstitial lung disease with subpleural reticulations and ground-glass opacities in the lower lobes. Patient in supine position. ***Bottom panel:*** Extensive lung fibrosis with coarse reticular honeycombing and traction bronchiectasis. *(Courtesy of Rishi Agrawal, Northwestern University.)*

indicated only in patients with atypical findings on chest imaging. The histologic pattern on lung biopsy may predict the risk of progression of ILD, with the NSIP pattern having a better prognosis than UIP.

**Pulmonary Arterial Hypertension** PAH results from vascular remodeling of small (<500 μm) pulmonary arteries. PAH develops in 8–12% of patients with SSc as a late complication and can be an isolated abnormality or associated with ILD. PAH is defined as a mean pulmonary artery pressure ≥20 mmHg with a pulmonary capillary wedge pressure ≤15 mmHg and pulmonary vascular resistance >3 Wood units. The natural history of SSc-associated PAH is variable but often follows a progressive course leading to right heart failure. The 3-year survival of SSc patients with untreated PAH is <50%. Risk factors include limited cutaneous disease, older age at disease onset, high numbers of cutaneous telangiectasia, and antibodies to centromere, U3-RNP (fibrillarin), and B23. Mutations in the *BMPR2* gene implicated in idiopathic PAH are not associated with SSc-PAH.

In SSc, PAH is often asymptomatic in early stages. Patients present with exertional dyspnea and reduced exercise capacity. With progression, angina, near-syncope, and symptoms and signs of right-sided heart failure appear. Physical examination may show tachypnea, a loud pulmonic component of the $S_2$ heart sound, pulmonic/tricuspid regurgitation murmur, palpable right ventricular heave, elevated jugular venous pressure, and dependent edema. Doppler echocardiography is a noninvasive screening method for estimating the pulmonary arterial pressure. In light of the poor prognosis of untreated PAH and better therapeutic response in patients with early diagnosis, SSc patients should be screened for PAH at initial evaluation and annually thereafter. Estimated pulmonary artery systolic pressure >40 mmHg at rest or tricuspid regurgitation jet velocities >3 m/s suggest PAH. Pulmonary function testing may show a reduced $DL_{CO}$ in isolation or out of proportion with the severity of restriction. Because echocardiography can over- or underestimate pulmonary artery pressures, cardiac catheterization is required to confirm the diagnosis of suspected PAH; assess its severity, including the degree of right heart dysfunction; rule out veno-occlusive disease and other cardiac (postcapillary) causes of pulmonary hypertension; and provide prognostic parameters. Distinguishing PAH in SSc patients from pulmonary hypertension secondary to pulmonary fibrosis and hypoxia can be difficult. Serum levels of N-terminal pro-brain natriuretic peptide (NT-proBNP) correlate with the presence and severity of PAH, as well as survival. While NT-proBNP can be useful in PAH screening and in monitoring the response to treatment, elevated levels are not specific for PAH and also occur in other forms of right and left heart disease. Despite more favorable hemodynamics of SSc-associated PAH, its prognosis is worse and treatment response poorer than for idiopathic PAH. This is most likely due to frequent concurrence of ILD and cardiac disease in patients.

## GASTROINTESTINAL INVOLVEMENT

Variable involvement of the gastrointestinal (GI) tract, which can affect any level, is the most common internal organ manifestation of SSc, seen in up to 90% of SSc patients with both limited and diffuse cutaneous disease (Table 360-6). Severe GI tract involvement is associated with male sex and specific autoantibodies. The pathologic findings of GI involvement in SSc include fibrosis, smooth-muscle atrophy, and obliterative small vessel vasculopathy throughout the length of the GI tract. Together they have major impact on quality of life, malnutrition, and mortality.

**Upper Gastrointestinal Tract Involvement** Decreased oral aperture interferes with regular dental hygiene. Loss of periodontal ligament attaching teeth to the alveolar bone leads to teeth loosening. Additional oropharyngeal manifestations due to a combination of xerostomia, shortened frenulum, and resorption of the mandibular condyles cause much distress. Most SSc patients show symptoms of GERD (heartburn, regurgitation, and dysphagia) due to a combination of reduced lower esophageal sphincter pressure resulting in reflux, impaired esophageal clearance of refluxed gastric contents due to diminished motility, and delayed gastric emptying. Calcium channel antagonists and phosphodiesterase inhibitors commonly used to treat Raynaud's phenomenon in SSc can further aggravate reflux. Esophageal manometry shows abnormal motility in most patients, even in the absence of symptoms. Common extraesophageal GERD manifestations in SSc include hoarseness, cough, and chronic microaspiration, which can aggravate underlying ILD. Characteristic chest CT findings include dilated patulous esophagus with intraluminal air. Esophageal dilation is associated with the severity of ILD. Endoscopy may be indicated in patients with dysphagia in order to rule out opportunistic infections with *Candida*, herpes virus, and cytomegalovirus. Severe erosive esophagitis may be found in patients with minimal symptoms. Esophageal strictures and Barrett's esophagus may complicate chronic GERD in SSc patients. Because Barrett's metaplasia is associated with increased risk of adenocarcinoma, these patients require regular surveillance endoscopy with biopsy.

Gastroparesis with early satiety, abdominal distention, and aggravated reflux symptoms is common. Barium contrast studies are neither sensitive nor specific for evaluation of gastric involvement in SSc. Gastric antral vascular ectasia (GAVE) in the antrum may occur. These subepithelial lesions, reflecting the widespread small vessel vasculopathy of SSc, are called "watermelon stomach" due to their endoscopic appearance. GAVE may present with recurrent episodes of GI bleeding, resulting in chronic unexplained anemia. Patients with SSc show increased prevalence of *Helicobacter pylori* infection in the stomach.

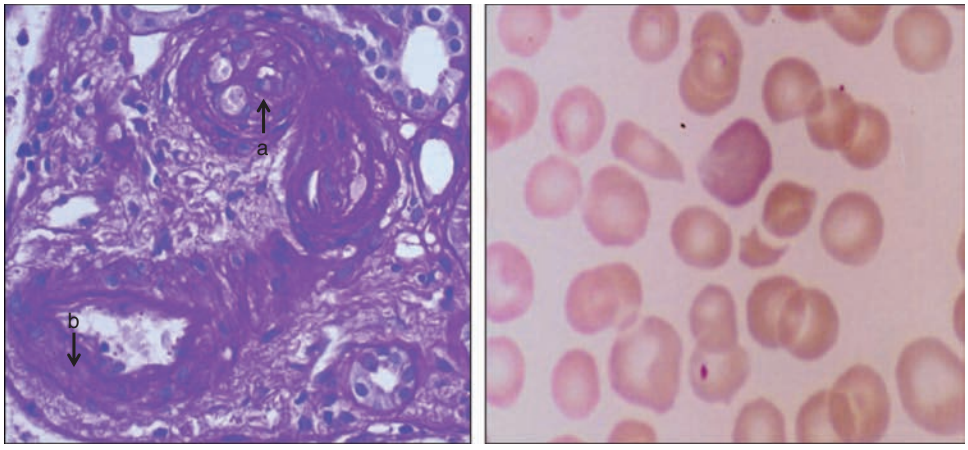
**Lower Gastrointestinal Tract and Anorectal Involvement** Weight loss and malnutrition due to a combination impaired intestinal motility, malabsorption, and chronic diarrhea secondary to bacterial overgrowth are common. Fat and protein malabsorption and vitamin $B_{12}$ and vitamin D deficiency ensue and may be further exacerbated by pancreatic insufficiency. Disturbed intestinal motor function can also lead to episodic intestinal pseudo- obstruction, with symptoms that are indistinguishable from those of delayed gastric emptying. Patients present with acute abdominal pain, nausea, and vomiting, and radiographic studies show acute intestinal obstruction. A major diagnostic challenge is differentiating pseudo-obstruction, which responds to supportive care and intravenous nutritional supplementation, from mechanical obstruction. Colonic involvement in SSc may result in constipation, occasionally complicated by sigmoid volvulus, fecal incontinence, GI bleeding from telangiectasia, and rectal prolapse. In late-stage SSc, wide-mouth sacculations or diverticula occur in the

**TABLE 360-6 Prominent Gastrointestinal Manifestations of Systemic Sclerosis and Their Management**

| SITE | PRINCIPAL MANIFESTATION | MANAGEMENT |
|---|---|---|
| Oropharynx | Diminished oral aperture<br>Dry mouth<br>Periodontitis, gingivitis<br>Swallowing difficulty | Periodontal care<br>Artificial saliva<br>Swallowing therapy |
| Esophagus | Reflux<br>Dysphagia<br>Strictures<br>Barret's metaplasia | Lifestyle modifications<br>Prokinetic drugs<br>Proton pump inhibitors<br>Endoscopic procedures |
| Stomach | Gastroparesis<br>Gastric antral vascular ectasia (GAVE, watermelon stomach) | Prokinetic agents<br>Endoscopic laser cryotherapy |
| Small and large intestines | Bacterial overgrowth (SIBO)<br>Diarrhea/constipation<br>Pseudo-obstruction<br>Pneumatosis intestinalis<br>Malabsorption<br>Colonic pseudodiverticula | Laxatives<br>Prokinetic agents<br>Rotating antibiotics<br>Octreotide<br>Parenteral nutritional support |
| Anorectum | Sphincter incompetence | Biofeedback, sacral nerve stimulation, surgery |

*Abbreviation:* SIBO, small intestinal bacterial overgrowth.

colon and occasionally cause perforation and bleeding. An unusual radiologic finding is pneumatosis cystoides intestinalis, which is due to air trapping in the bowel wall that may rarely rupture and cause benign pneumoperitoneum. Studies of the fecal microbiota in SSc show a reduction in protective butyrate-producing bacteria, potentially promoting a proinflammatory intestinal microenvironment. Although the liver is rarely affected in patients with SSc, primary biliary cirrhosis may occur, particularly in patients with limited cutaneous disease.




*A* *B*

**FIGURE 360-13 Biopsy and hematologic findings in scleroderma renal crisis.** ***A.*** Renal biopsy demonstrating intimal proliferation and myxoid changes in medium-sized renal arteries. ***B.*** Fragmentation of red blood cells due to intravascular hemolysis. *(Courtesy of Drs. Edward Stern and Christopher Denton, Royal Free Hospital, London, UK.)*

## RENAL INVOLVEMENT: SCLERODERMA RENAL CRISIS

Scleroderma renal crisis presents with accelerated hypertension accompanied by acute kidney injury and progressive failure. This acute life-threatening complication occurs in <15% of SSc patients, almost always within 4 years of disease onset. Scleroderma renal crisis can occasionally even be the presenting manifestation of SSc. While short-term survival in scleroderma renal crisis was <10% prior to the advent of angiotensin-converting enzyme (ACE) inhibitors, outcomes for this serous complication have shown great improvement. The pathogenesis involves obliterative vasculopathy of the renal arcuate and interlobular arteries, with consequent intravascular hemolysis **(Fig. 360-13)**. Progressive reduction in renal blood flow, aggravated by vasospasm, leads to increased juxtaglomerular renin secretion and angiotensin II generation, with further renal vasoconstriction resulting in a vicious cycle that culminates in accelerated hypertension. Risk factors for scleroderma renal crisis include African-American race, male sex, and diffuse or progressive skin involvement. Up to 50% of patients with scleroderma renal crisis have antibodies to anti-RNA polymerase III, whereas patients with anti-centromere antibodies appear protected. Palpable tendon friction rubs, pericardial effusion, new unexplained anemia, and thrombocytopenia may be harbingers of impending scleroderma renal crisis. High-risk patients with early-stage SSc should monitor their blood pressure daily. Because glucocorticoid use is associated with scleroderma renal crisis, prednisone in high-risk SSc patients should be taken only when absolutely required and at low doses (<10 mg/d).

Patients with scleroderma renal crisis characteristically present with accelerated hypertension (generally >150/90 mmHg) and progressive oliguric renal insufficiency. However, ~10% of patients present with blood pressure in the normal range. Normotensive renal crisis is generally associated with a poor outcome. Headache, blurred vision, congestive heart failure, and pulmonary edema may accompany elevation of blood pressure. Moderate thrombocytopenia and microangiopathic hemolysis with fragmented red blood cells can be seen, and urinalysis typically shows mild proteinuria, granular casts, and microscopic hematuria. Progressive oliguric renal failure over several days generally follows. Scleroderma renal crisis is sometimes misdiagnosed as thrombotic thrombocytopenic purpura (TTP) or other forms of thrombotic microangiopathy. In such cases, renal biopsy and determination of serum von Willebrand factor–cleaving protease activity may be of benefit. The presence of oliguria or a creatinine >3 mg/dL at initial presentation predicts poor outcome (permanent hemodialysis and mortality), as do vascular thrombosis and glomerular ischemic collapse on renal biopsy. Crescentic glomerulonephritis in the setting of SSc has been described and may be associated with myeloperoxidase-specific antineutrophil cytoplasmic antibodies (ANCAs). Membranous glomerulonephritis may occur in patients treated with D-penicillamine. Asymptomatic renal impairment can occur in up to half of SSc patients, is commonly associated with other vascular manifestations, and rarely progresses to end-stage renal failure.

## CARDIAC INVOLVEMENT

Although it is often silent, cardiac involvement is detected in 10–50% of SSc patients screened with sensitive diagnostic tools. Clinical cardiac involvement is more frequent in dcSSc than in lcSSc and may be primary or secondary to PAH, ILD, or renal involvement. Cardiac involvement in SSc is associated with poor outcomes. The endocardium, myocardium, and pericardium may each be affected separately or together. Pericardial involvement is manifested as pericarditis, pericardial effusions, constrictive pericarditis, and rarely, cardiac tamponade. Conduction system fibrosis is common and may be silent or manifested by heart block. Other arrhythmias include premature ventricular contractions, atrial fibrillation, and supraventricular tachycardia. Microvascular involvement, recurrent vasospasm, and ischemia-reperfusion injury contribute to patchy myocardial fibrosis, and the resulting systolic or diastolic left ventricular dysfunction may progress to overt heart failure. Acute or subacute myocarditis leading to left ventricular dysfunction may occur, best diagnosed by cardiac magnetic resonance imaging or endomyocardial biopsy. While conventional echocardiography has low sensitivity for detecting preclinical heart involvement in SSc, newer modalities such as tissue Doppler echocardiography (TDE), cardiac MRI, and nuclear imaging (single-photon emission CT [SPECT]) reveal a high prevalence of abnormal myocardial function or perfusion. The serum levels of NT-proBNP, a ventricular hormone elevated in SSc-PAH, may also have utility as a marker of primary cardiac involvement.

## MUSCULOSKELETAL COMPLICATIONS

Musculoskeletal complications are commonly seen in SSc. Carpal tunnel syndrome may be a presenting disease manifestation, and generalized arthralgia and stiffness are prominent in early disease. Mobility of both small and large joints is progressively impaired, and fixed contractures at the proximal interphalangeal joints and wrists ensue. Large joint contractures occur in patients with dcSSc and are frequently accompanied by tendon friction rubs. These are characterized by coarse leathery crepitation heard or palpated upon passive joint movement and are caused by fibrosis and adhesion of the tendon sheaths and fascial planes at the affected joint. The presence of tendon friction rubs is associated with increased risk for renal and cardiac complications and reduced survival. Synovitis seen on ultrasound or MRI is common; occasional SSc patients develop erosive polyarthritis in the hands, and some may have a seropositive rheumatoid arthritis overlap. Muscle weakness is common and multifactorial; deconditioning, disuse

atrophy, malnutrition, inflammation, and fibrosis may all contribute. A chronic noninflammatory form of myopathy characterized by atrophy and fibrosis with mildly elevated muscle enzymes occurs in late-stage SSc. Bone resorption in the terminal phalanges causes the characteristic loss of the distal tufts (acro-osteolysis) (Fig. 360-5). Resorption of the mandibular condyles can lead to bite difficulties. Osteolysis can also affect the ribs and distal clavicles.

### LESS RECOGNIZED DISEASE MANIFESTATIONS

Dry eyes and dry mouth (sicca complex) are common in SSc. Biopsy of the minor salivary glands in these cases shows fibrosis rather than focal lymphocytic infiltration characteristic of primary Sjögren's syndrome **(Chap. 361)**. Hypothyroidism resulting from Graves' or Hashimoto's disease is common, particularly in lcSSc, and may be underrecognized. Whereas the central nervous system is generally spared in SSc, unilateral or bilateral sensory trigeminal neuropathy can occur. Erectile dysfunction is a frequent and occasionally initial disease manifestation. Inability to attain or maintain penile erection is due to vascular insufficiency and fibrosis of corporeal smooth muscle and responds poorly to medical therapy. Sexual performance is also adversely affected in women. While fertility is not impaired in SSc, pregnancy is associated with a higher risk of adverse fetal outcomes. Furthermore, cardiopulmonary involvement may worsen during pregnancy, and new onset of scleroderma renal crisis has been described.

**Cancer** Epidemiologic studies indicate an increased cancer risk in SSc. Lung cancer and esophageal adenocarcinoma typically occur in the setting of long-standing ILD or GERD and may be linked to chronic inflammation and tissue repair. In contrast, breast, lung, and ovarian carcinomas and lymphomas tend to occur in close temporal association with the onset of SSc, particularly in patients who have autoantibodies to RNA polymerase III. In this scenario, SSc may represent a paraneoplastic syndrome that is triggered by the antitumor immune response.

### LABORATORY EVALUATION AND BIOMARKERS

Mild microcytic anemia is frequent and may indicate recurrent GI bleeding caused by GAVE or chronic esophagitis. Macrocytic anemia may be caused by folate and vitamin $B_{12}$ deficiency due to small-bowel bacterial overgrowth and malabsorption or by drugs such as methotrexate. Microangiopathic hemolytic anemia caused by mechanical fragmentation of red blood cells during their passage through microvessels coated with fibrin or platelet thrombi is a hallmark of scleroderma renal crisis. The erythrocyte sedimentation rate (ESR) is generally normal in SSc; an elevation may signal coexisting myositis or malignancy.

Antinuclear autoantibodies are detected in almost all patients with SSc. Anti-topoisomerase I (Scl-70) and anti-centromere antibodies are mutually exclusive and highly specific for SSc. Topoisomerase I (Scl-70) antibodies are associated with increased risk of ILD and poor outcomes. Anti-centromere antibodies are associated with PAH, but only infrequently with significant cardiac, pulmonary, or renal involvement. Nucleolar immunofluorescence pattern may indicate antibodies to U3-RNP (fibrillarin), Th/To, or PM/Scl, whereas speckled immunofluorescence indicates antibodies to RNA polymerase III, associated with increased risk of scleroderma renal crisis and malignancy **(Fig. 360-14)**.

### DIAGNOSIS, STAGING, AND MONITORING

The diagnosis of SSc is made primarily on clinical grounds and is generally straightforward in patients with established disease. Diagnostic criteria developed for classification are >90% specific and sensitive for SSc. The presence of skin induration with a characteristic symmetric distribution pattern associated with typical visceral organ manifestations establishes the diagnosis with a high degree of certainty. In lcSSc, a history of Raynaud's phenomenon and GERD symptoms, coupled with sclerodactyly and nailfold capillary changes, often in combinations with cutaneous telangiectasia and calcinosis cutis, helps to establish the diagnosis. Primary Raynaud's disease is a benign condition that must be differentiated from early or limited SSc. Nailfold microscopy is particularly helpful in this situation, because in contrast to SSc, nailfold capillaries are normal. Diagnosing SSc at an early stage may be a challenge. In dcSSc, initial symptoms are often nonspecific, Raynaud's phenomenon may be absent, and physical examination may only show upper extremity edema and puffy fingers. Early-stage SSc might be initially misdiagnosed as arthritis, SLE, myositis, or, most commonly, undifferentiated connective tissue disease leading to delays in diagnosis. Within weeks to months, Raynaud's phenomenon and advancing skin induration appear. SSc-specific autoantibodies provide a high degree of diagnostic certainty. Raynaud's phenomenon with fingertip ulcerations or other evidence of digital ischemia, coupled with telangiectasia, distal esophageal dysmotility, unexplained ILD or PAH, or accelerated hypertension with renal failure in the absence of clinically evident skin induration, suggests the diagnosis of SSc *sine* scleroderma. Patients with a new diagnosis of SSc should be screened for ILD, followed by regular pulmonary monitoring for several years **(Fig. 360-15)**.

## MANAGEMENT OF SYSTEMIC SCLEROSIS

### OVERVIEW: GENERAL PRINCIPLES

To date, no therapy has been shown to significantly alter the natural history of SSc. In contrast, numerous interventions are effective in alleviating the symptoms, slowing the progression of the cumulative organ damage, and reducing disability. Moreover, a significant decrease in disease-related mortality has been noted during the past 25 years. In light of the marked heterogeneity in disease manifestations, course, and outcomes, optimized care for patients with SSc requires a thoughtful "precision medicine" approach that is tailored specifically to each individual patient's unique needs.

The following general principles should guide management **(Table 360-7)**: prompt and accurate diagnosis; patient subclassification and risk stratification based on clinical, radiologic, and laboratory evaluation, including autoantibody profiles and prognostic and predictive biomarkers; early recognition of organ-based complications and assessment of their extent, severity, and likelihood of deterioration; regular monitoring for disease progression, new complications, and response to and side effects of therapy; adjusting therapy; and patient education. In order to minimize the risk of irreversible organ damage, management should be individualized and proactive, with regular screening and initiation of appropriate intervention at the earliest possible opportunity. In light of the complex and multisystemic nature of the SSc, a team-oriented management approach integrating appropriate specialists should be pursued. Most patients are treated with a combination of drugs that impact different aspects of the disease. Patients should be encouraged to become familiar with potential complications and therapeutic options, including interventional trials, and natural history and should be empowered to partner with their treating physicians. This requires a long-term relationship between patient and physician, with ongoing counseling, encouragement, and two-way dialogue.

### DISEASE-MODIFYING THERAPY: IMMUNOSUPPRESSIVE AGENTS

Immunosuppressive agents used in other autoimmune diseases have generally shown modest or no benefit in SSc. Glucocorticoids alleviate stiffness and aching in early inflammatory-stage dcSSc but do not influence the progression of skin or internal organ involvement. Since their use is associated with an increased risk of scleroderma renal crisis, glucocorticoids should be given only when absolutely necessary, at the lowest dose possible, and for brief periods only.

Cyclophosphamide has been extensively studied in light of its efficacy in the treatment of vasculitis **(Chap. 363)**, SLE **(Chap. 356)**, and other autoimmune diseases **(Chap. 355)**. Both oral and intravenous cyclophosphamide have been shown to reduce the progression of SSc-associated ILD, with stabilization and, rarely, modest improvement

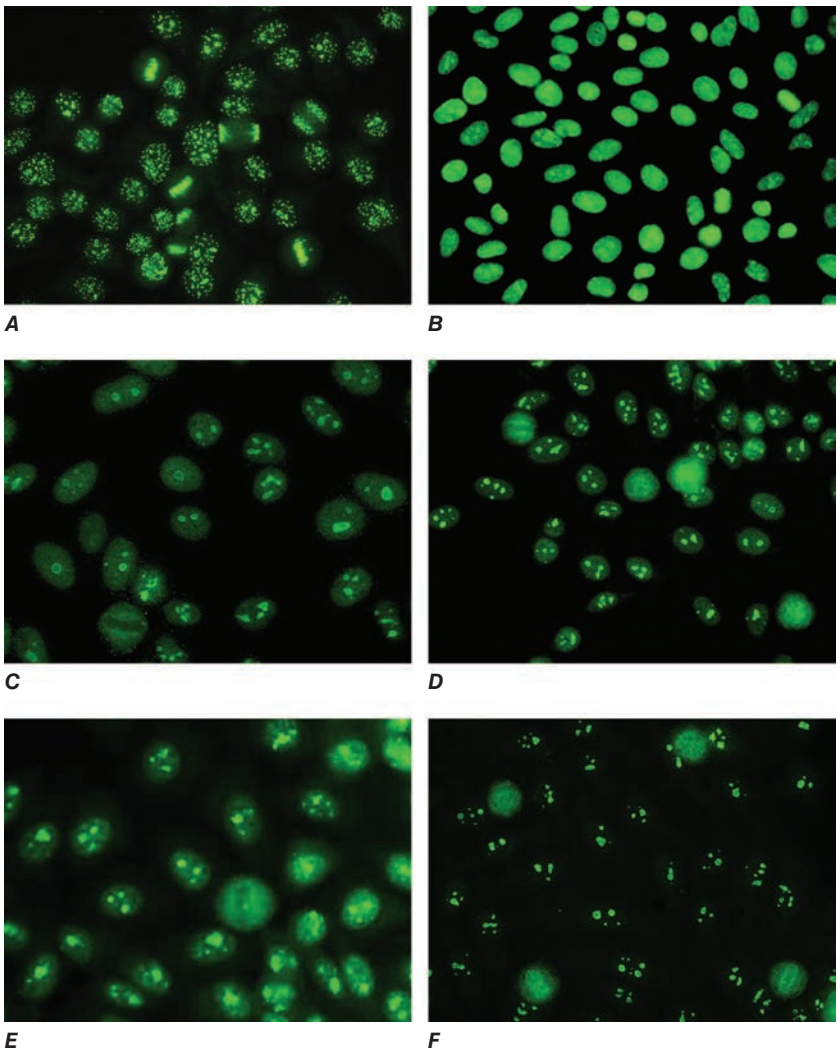



**FIGURE 360-14 SSc-associated autoantibodies: immunofluorescence.** Indirect immunofluorescence of SSc serum samples using HEp-2 substrate. Representative images show (***A***) anti-centromere; (***B***) anti-Scl-70/topoisomerase I; (***C***) anti-PM/Scl; (***D***) anti-Th/To; (***E***) anti-RNA polymerase III; and (***F***) anti-fibrillarin/U3RNP antibodies. Variations in immunostaining are clues to autoantibody specificity, but immunoassays with purified autoantigens are needed to confirm antigen specificity. *(Courtesy of Marvin Fritzler and Susan Copple, Inova Diagnostics Inc., San Diego, CA.)*

of pulmonary function, HRCT findings, respiratory symptoms, and skin induration. These benefits of cyclophosphamide need to be balanced against its potential toxicity, including bone marrow suppression, opportunistic infections, hemorrhagic cystitis and bladder cancer, premature ovarian failure, and late secondary malignancies.

Methotrexate had modest effect on SSc skin involvement in small studies. Mycophenolate mofetil was evaluated in both open-label and randomized controlled trials. Both skin induration and ILD improved in patients treated with mycophenolate mofetil, and the drug was well tolerated. Tocilizumab, a monoclonal antibody that blocks IL-6 receptor signaling, also showed benefit on both skin and lung involvement in randomized SSc trials. Open-label studies and small clinical trials provide some support for rituximab, a monoclonal antibody directed against the mature B-cell marker CD20, along with extracorporeal photopheresis, IV immunoglobulin, and abatacept, a fusion protein that inhibits T-cell co-stimulation and function. The use of cyclosporine, azathioprine, hydroxychloroquine (Plaquenil), thalidomide, and rapamycin for SSc therapy is currently not well supported by the literature. Intensive immune ablation using high-dose chemotherapy, followed by autologous hematopoietic stem cell reconstitution therapy (HSCT), was associated with durable remission and improved long-term survival in multiple randomized clinical trials. Currently, HSCT is indicated for selected patients with severe SSc but carries potential morbidity and mortality, as well as significant cost. Additional forms of potentially disease-modifying cellular therapies are under investigation.

**Antifibrotic Therapy** Because tissue fibrosis underlies organ damage in SSc, drugs that interfere with the fibrotic process represent a rational therapeutic approach. In older retrospective studies, D-penicillamine was shown to stabilize skin induration, prevent new internal organ involvement, and improve survival. However, a randomized controlled clinical trial in early active SSc found no difference in the extent of skin involvement between patients treated with standard-dose (750 mg/d) or very low-dose (125 mg every other day) D-penicillamine. Recent clinical trials show benefit of the tyrosine kinase inhibitor nintedanib, alone or in combination with mycophenolate, in patients with SSc-ILD, with significant slowing of the loss of lung function.

**Vascular Therapy** The goal of Raynaud's therapy is to control episodes, prevent and enhance the healing of ischemic complications, and slow the progression of obliterative vasculopathy. Patients should dress warmly, minimize cold exposure, and avoid drugs that precipitate or exacerbate vasospastic episodes. Extended-release dihydropyridine calcium channel blockers such as amlodipine and nifedipine, as well as diltiazem, ameliorate Raynaud's phenomenon, but their use is often limited by side effects (palpitations, dependent edema, worsening gastroesophageal reflux). While ACE inhibitors do not reduce the

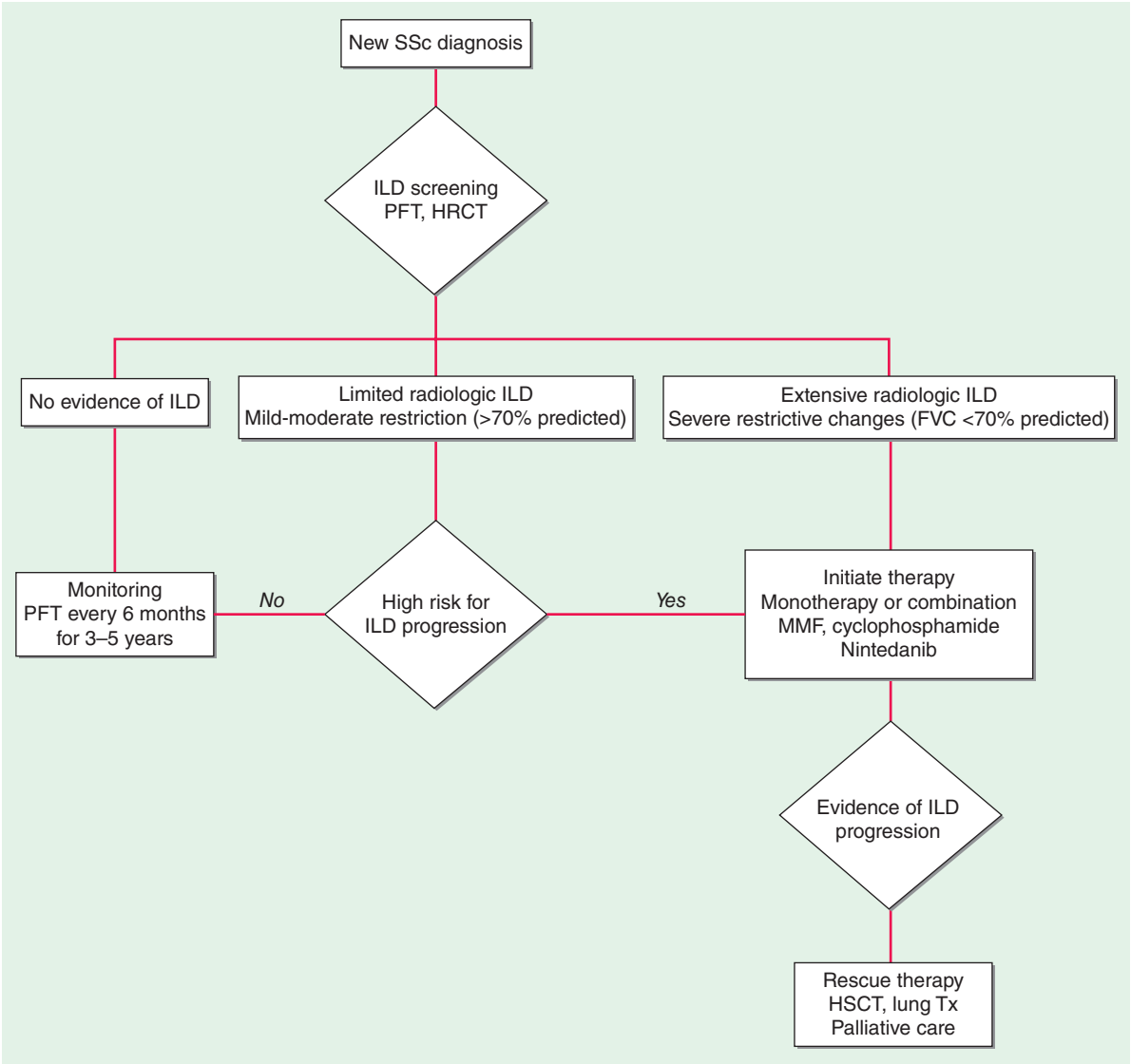



**FIGURE 360-15 Proposed algorithm for screening, monitoring, and treatment of systemic sclerosis (SSc)-associated interstitial lung disease (ILD).** FVC, forced vital capacity; HRCT, high-resolution computed tomography; HSCT, hematopoietic stem cell transplantation; MMF, mycophenolate mofetil; PFT, pulmonary function testing; Tx, transplantation. *(Adapted from A Perelas et al: Lancet Respir Med 8:304, 2020.)*

frequency or severity of episodes, angiotensin II receptor blockers such as losartan are effective and well tolerated. Patients with Raynaud's phenomenon unresponsive to these therapies may require the addition of $\alpha_1$-adrenergic receptor blockers (e.g., prazosin), phosphodiesterase-5 inhibitors (e.g., sildenafil), topical nitroglycerine, and intermittent IV infusions of prostaglandins. Low-dose aspirin and dipyridamole prevent platelet aggregation and may have a role as adjunctive agents. In patients with ischemic digital tip ulcerations, the endothelin-1 receptor antagonist bosentan reduces the risk of new ulcers. Digital sympathectomy and intradigital injections of botulinum type A (Botox) may be considered in patients with severe ongoing ischemia. Empirical long-term therapy with statins and antioxidants may retard the progression of vascular damage and obliteration. There is limited evidence-based information for the treatment of cardiac complications of SSc, which should be guided by specialists experienced in their diagnosis and management. While selective beta blockers such as metoprolol can precipitate vasospasm, nondihydropyridine calcium channel blockers can be used for rate control in atrial arrhythmias, and nonselective alpha/beta blockers such as carvedilol can be used for improving myocardial perfusion and left ventricular systolic function.

**TABLE 360-7 Key Principles in Management**

- Establish early and accurate diagnosis.
- Detect and evaluate internal organ involvement.
- Define clinical disease stage and activity.
- Tailor individualized therapy to each patient's unique needs.
- Assess treatment response, and adjust therapy as needed; monitor for disease activity, progression, and new complications.

## TREATMENT

### SSC-ASSOCIATED INTERSTITIAL LUNG DISEASE

ILD is a leading cause of death in patients with SSc. However, because the course of SSc-associated ILD is highly variable, it is important to identify patients who are at high risk for disease progression. The extent of ILD on HRCT and the FVC at initial evaluation and the decline in FVC during the preceding 12-month period are helpful in identifying these patients. Additionally, male sex, older age at initial presentation, progressive skin involvement, and myocardial disease may be risk factors for ILD progression. Patients at high risk for ILD should be monitored by performing PFTs every 6 months (Fig. 360-15); serial HRCT imaging is not recommended. Cyclophosphamide, given IV or orally for 6–12 months,

and mycophenolate mofetil slow the decline in lung function and improve respiratory symptoms, but cyclophosphamide is associated with more frequent side effects. The efficacy and optimal duration of antifibrotic therapy with nintedanib, which was recently approved for SSc-associated ILD, are currently under investigation. In patients who show continued progression of ILD despite medical therapy, lung transplantation might be considered as a life-prolonging procedure, although significant GERD contributing to organ rejection is a concern in SSc. Recurrence of SSc-ILD in transplanted lung allografts has not been reported.

### TREATMENT OF GASTROINTESTINAL COMPLICATIONS

Because oral problems, including decreased oral aperture, decreased saliva production, gum recession, periodontal disease, and teeth loss, are common, regular dental care is recommended. Gastroesophageal reflux is very common in SSc and may occur in the absence of symptoms. Patients should be instructed to elevate the head of the bed, eat frequent small meals, and avoid alcohol, caffeine, known reflux exacerbants, and meals before bedtime. Proton pump inhibitors to reduce acid reflux may need to be given in relatively high doses. Prokinetic agents such as metoclopramide, erythromycin (a motilin agonist), and domperidone may occasionally be helpful in SSc but are frequently associated with side effects. Botulinum toxin injection sometimes ameliorates impaired gastric emptying. Antireflux procedures such as Nissen fundoplication can result in secondary achalasia and generally should be avoided. Episodic bleeding from GAVE (watermelon stomach) may be treated with endoscopic ablation using laser or argon plasma photocoagulation, although it commonly recurs. In some patients, enteral feeding and/or decompression via percutaneous gastrostomy or jejunostomy may become necessary. Small intestinal bacterial overgrowth secondary to gut dysmotility causes abdominal bloating and diarrhea and may lead to malabsorption and severe malnutrition. Short courses of rotating broad-spectrum antibiotics such as metronidazole, erythromycin, and rifaximin can eradicate bacterial overgrowth. Small bowel hypomotility may respond to octreotide, but pseudo-obstruction is difficult to treat. Fecal incontinence, a frequent but underreported complication, may respond to antidiarrheal medication, biofeedback, sphincter augmentation, and sacral neuromodulation. Potential malnutrition should be routinely assessed.

### TREATMENT OF PULMONARY ARTERIAL HYPERTENSION

In SSc, PAH carries an extremely poor prognosis and accounts for 30% of deaths. Because PAH is asymptomatic until advanced, patients with SSc should be screened at initial evaluation and regularly thereafter. Treatment is generally started with an oral endothelin-1 receptor antagonist such as bosentan or a phosphodiesterase-5 inhibitor such as sildenafil. Recently, the soluble guanylate cyclase stimulator riociguat, which acts by increasing the production of nitric oxide, and the selective IP prostacyclin receptor agonist selexipag were shown to improve PAH symptoms and survival. Patients may also require diuretics and digoxin. If hypoxemia is documented, supplemental oxygen should be prescribed in order to avoid secondary pulmonary vasoconstriction. Prostacyclin analogues such as epoprostenol or treprostinil can be given by continuous IV or SC infusion or via intermittent nebulized inhalations. Combination therapy with different classes of agents acting additively or synergistically is often necessary. Lung transplantation remains an option for selected SSc patients with PAH who fail medical therapy, and 2-year survival rates (64%) are comparable to those of idiopathic ILD or PAH.

### MANAGEMENT OF RENAL CRISIS

Scleroderma renal crisis represents a medical emergency. Since its outcome is largely determined by the extent of renal damage at the time that aggressive therapy is initiated, prompt recognition of impending or early scleroderma renal crisis is essential, and efforts should be made to avoid its occurrence. High-risk SSc patients (early disease, extensive and progressive skin involvement, tendon friction rubs, and anti-RNA polymerase III antibodies) should be instructed to monitor their blood pressure daily and report significant alterations immediately. Potentially nephrotoxic drugs should be avoided, and glucocorticoids should be used only when absolutely necessary and at low doses. Patients presenting with scleroderma renal crisis should be immediately hospitalized. Once other causes of acute renal disease are excluded, treatment should be started promptly with titration of short-acting ACE inhibitors, with the goal of rapid normalization of the blood pressure. In patients with persistent hypertension, addition of angiotensin II receptor blockers, calcium channel blockers, endothelin-1 receptor blockers, prostacyclins, and direct renin inhibitors should be considered. In light of evidence for intrarenal complement pathway activation in some patients with scleroderma renal crisis, addition of eculizumab to ACE inhibitors may be considered. Up to two-thirds of patients with scleroderma renal crisis necessitate dialysis. Substantial renal recovery can occur following an episode of scleroderma renal crisis, and renal replacement therapy can be ultimately discontinued in 30–50% of patients. Kidney transplantation is appropriate for patients unable to discontinue dialysis after 2 years. Survival of transplanted SSc patients is comparable to that of other diseases, and recurrence of renal crisis is rare.

### SKIN CARE

Because skin involvement in SSc is never life-threatening and stabilizes, and may even regress spontaneously, disease management should not be dictated by its cutaneous manifestations. The inflammatory symptoms of early skin involvement can be controlled with antihistamines and short-term use of low-dose glucocorticoids (<5 mg/d of prednisone). Cyclophosphamide and methotrexate have modest effects on skin induration. Because the skin is dry, the use of hydrophilic ointments and bath oils is encouraged, and regular skin massage is helpful. Telangiectasia, which presents a cosmetic problem, especially on the face, can be treated with pulsed dye laser. Ischemic digital ulcerations should be protected by occlusive dressings to promote healing and prevent infection. Infected skin ulcers are treated with topical antibiotics and surgical debridement. While no therapy has been shown to be effective in preventing soft-tissue calcific deposition or promoting its dissolution, reports support the use of minocycline, bisphosphonates, and topical or IV sodium thiosulfate (STS). Additional approaches include carbon dioxide laser treatment, extracorporeal shock-wave lithotripsy, and surgical high-speed microdrilling.

### TREATMENT OF MUSCULOSKELETAL COMPLICATIONS

Arthralgia and joint stiffness are very common and distressing manifestations in early-stage disease. Short courses of nonsteroidal anti-inflammatory agents, methotrexate, and cautious use of low-dose glucocorticoids alleviate symptoms. Physical and occupational therapy can be effective for preventing loss of musculoskeletal function and joint contractures and should be initiated early.

## COURSE

The natural history of SSc is highly variable and difficult to predict, especially in early stages of the disease. Patient with dcSSc tend to have a more rapidly progressive course and worse prognosis than those with lcSSc. Inflammatory symptoms of early dcSSc, such as fatigue, edema, joint pain, and pruritus, subside, and skin thickening reaches a plateau at 2–4 years after disease onset. It is during the early edematous/inflammatory stage that life-threatening visceral organ involvement may develop. While existing visceral organ involvement, such as ILD, may progress even after skin involvement peaks, new organ involvement is rare. Scleroderma renal crisis generally occurs within the first 4 years of disease. In late-stage disease (>6 years), the skin is usually soft and atrophic. Skin regression characteristically occurs in an order that is the reverse of initial involvement, with softening on the trunks followed by proximal and finally distal extremities; however, sclerodactyly and

fixed finger contractures generally persist. Relapse or recurrence of skin thickening after peak skin involvement has been reached is uncommon. Patients with lcSSc follow a clinical course that is markedly different than that of dcSSc. Raynaud's phenomenon typically precedes other disease manifestations by years or even decades. Visceral organ complications such as PAH generally develop late and progress slowly.

## PROGNOSIS

SSc confers a substantial increase in the risk of premature death. Age- and gender-adjusted mortality rates are five- to eightfold higher compared to the general population, and more than half of all patients with SSc die from their disease. In one population-based study of SSc, the median survival was 11 years. In patients with dcSSc, 5- and 10-year survival rates are 70% and 55%, respectively, whereas in patients with lcSSc, 5- and 10-year survival rates are 90% and 75%, respectively. The prognosis correlates with the extent of skin involvement, which itself is a surrogate for visceral organ involvement. Major causes of death are PAH, pulmonary fibrosis, GI involvement, and cardiac disease. Scleroderma renal crisis is associated with a 30% 3-year mortality. Lung cancer and excess cardiovascular deaths also contribute to increased mortality. Markers of poor prognosis include male gender, African-American race, older age at disease onset, extensive skin thickening with truncal involvement, palpable tendon friction rubs, and evidence of significant or progressive visceral organ involvement. Laboratory predictors of increased mortality at initial evaluation include an elevated ESR, anemia, proteinuria, and anti–topoisomerase I (Scl-70) antibodies. In one study, SSc patients with extensive skin involvement, lung vital capacity <55% predicted, significant GI involvement (pseudo-obstruction or malabsorption), clinical cardiac involvement, or scleroderma renal crisis had a 9-year survival <40%. The severity of PAH predicts mortality, and patients with mean pulmonary arterial pressure ≥45 mmHg had a 33% 3-year survival. The advent of ACE inhibitors in scleroderma renal crisis had a dramatic impact on survival, increasing survival from <10% at 1 year in the pre–ACE inhibitor era to >70% 3-year survival at the present time. Moreover, 10-year survival in SSc improved from <60% in the 1970s to >66–78% in the 1990s, a trend that reflects both earlier detection and better management of complications.

## LOCALIZED SCLERODERMA

The term *scleroderma* describes a group of localized skin disorders (Table 360-1). These occur more commonly in children than in adults and, in marked contrast to SSc, are generally not complicated by Raynaud's phenomenon or significant internal organ involvement. Morphea presents as solitary or multiple circular patches of thick skin or, rarely, as widespread induration (generalized or pansclerotic morphea); the fingers are generally spared. Linear scleroderma may affect subcutaneous tissues, leading to fibrosis and atrophy of supporting structures, tendons, muscle, and even bone. In children, the growth of affected long bones can be retarded. When linear scleroderma crosses large joints, significant contractures can develop.

## MIXED CONNECTIVE TISSUE DISEASE

Patients who have lcSSc coexisting with features of SLE, polymyositis, and rheumatoid arthritis may have mixed connective tissue disease (MCTD). This overlap syndrome is generally associated with the presence of high titers of autoantibodies to U1-RNP. The characteristic initial presentation is Raynaud's phenomenon associated with puffy fingers and myalgia. Over time, sclerodactyly, soft-tissue calcinosis, and cutaneous telangiectasia may appear. Skin rash suggestive of SLE (malar erythema, photosensitivity) or dermatomyositis (heliotrope rash on the eyelids, erythematous rash on knuckles) occurs. Arthralgia is common, and some patients develop erosive polyarthritis. Pulmonary fibrosis and isolated or secondary PAH may develop. Other manifestations include esophageal dysmotility, pericarditis, Sjögren's syndrome, and renal disease, especially membranous glomerulonephritis. Laboratory evaluation shows elevated ESR and hypergammaglobulinemia. While anti-U1RNP antibodies are detected in high titers, SSc-specific autoantibodies are absent. In contrast to SSc, MCTD often responds to glucocorticoids, and the long-term prognosis is better than that of SSc. Whether MCTD is truly a distinct entity or is a subset of SLE or SSc remains controversial.

## EOSINOPHILIC FASCIITIS (DIFFUSE FASCIITIS WITH EOSINOPHILIA)

Eosinophilic fasciitis is a rare idiopathic disorder of adults associated with abrupt skin induration. The skin characteristically shows a coarse cobblestone "peau d'orange" appearance. In contrast to SSc, Raynaud's phenomenon and SSc-associated internal organ involvement and autoantibodies are absent. Furthermore, skin involvement spares the fingers. Full-thickness biopsy of the lesional skin reveals fibrosis of the subcutaneous fascia, with variable inflammation and eosinophil infiltration. In the acute phase of the illness, peripheral blood eosinophilia may be prominent. MRI appears to be a sensitive tool for the diagnosis of eosinophilic fasciitis. Eosinophilic fasciitis can occur in association with, or preceding, various myelodysplastic syndromes or multiple myeloma. Although glucocorticoids cause prompt resolution of eosinophilia, the skin shows slow and variable improvement. The prognosis of patients with eosinophilic fasciitis is generally good.

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# 361 Sjögren's Syndrome

**Haralampos M. Moutsopoulos, Clio P. Mavragani**

## DEFINITION, INCIDENCE, AND PREVALENCE

Sjögren's syndrome is a prototype autoimmune disease characterized by lymphocytic infiltration of the exocrine glands resulting in xerostomia, dry eyes (keratoconjunctivitis sicca), and profound B-cell hyperactivity. The syndrome has unique features since it presents with a wide clinical spectrum from organ-specific to systemic disease; can occur alone or in association with other systemic rheumatic diseases, more commonly rheumatoid arthritis, limited scleroderma, and systemic lupus erythematosus; and displays high probability of the development of lymphoma. Because of all these characteristics, it is an ideal model to study not only autoimmunity but also lymphoid malignancy.

Middle-aged women (female-to-male ratio 10-20:1) are primarily affected, although Sjögren's syndrome may occur at any age, including childhood. Patients with earlier disease onset express a more aggressive disease phenotype manifested by a high occurrence of systemic manifestations and serum autoantibodies. The prevalence of Sjögren's syndrome is ~0.5–1%, while 5–20% of patients with other autoimmune diseases can express sicca manifestations.